

# On zeros of gradients of harmonic functions on the upper half space.

Master's Thesis

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# Abstract

This document was prepared as a partial requirement to obtain the master's degree at IMPA, and it's based on existing literature. No original results are presented. This master thesis investigates the boundary behavior of harmonic functions on the upper half-space  $\mathcal{H}^d$ , focusing specifically on the conditions under which the gradient of a harmonic function can vanish on the boundary. The foundational motivation for this work stems from Electrical Impedance Tomography (EIT) and the classical Calderón problem, an inverse boundary value problem that aims to determine the internal conductivity of a body from voltage and current measurements taken at its boundary. While uniqueness for the Calderón problem is well established under general regularity conditions, the stability of the reconstruction remains severely ill-posed. A survey on the Calderón problem is presented at Chapter 1.

To explore the analytical mechanisms driving this instability, we study a striking phenomenon demonstrated by Bourgain and Wolff, who constructed a non-trivial harmonic function of class  $C^1$  up to the boundary whose trace and normal derivative vanish simultaneously on a boundary subset of positive measure. This form of "boundary invisibility" provides deep structural intuition for the severe logarithmic instability and the challenges inherent in partial data inverse problems.

This work systematically details the dimensional dichotomy of this phenomenon. In Chapter 2 we explore the two-dimensional setting ( $d = 2$ ), such boundary degeneracy is impossible; we provide a complex-analytic proof utilizing outer functions and the structural theory of Hardy spaces  $H^p$  to demonstrate that if the Dirichlet and Neumann data vanish concurrently on a set of positive measure, the function must be identically zero. Finally at Chapter 3 we explore the problem for higher dimensions ( $d \geq 3$ ), the phenomenon is realized through sophisticated tools in harmonic analysis. We detail the construction of a non-trivial harmonic function with vanishing boundary gradients in higher dimensions by deploying Poisson kernels, Riesz transforms, and Calderón-Zygmund decompositions, orchestrated via an inductive algorithm centered on highly localized correction functions.

**Keywords:** Calderón Problem, Electrical Impedance Tomography, Harmonic Functions, Hardy Spaces, Riesz Transforms.

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# Introduction

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We motivate this work by means of Electrical Impedance Tomography and the Calderón Problem.

Electrical Impedance Tomography (EIT) is a imaging technique that aims to recover the internal electrical properties of a body (typically conductivity or resistivity) from measurements taken at its boundary. The problem arises from the fact that different materials conduct electricity differently, and by injecting currents and measuring the resulting voltages at the boundary, one can attempt to infer information about the interior structure. This has applications ranging from medical imaging such as detecting anomalies in human tissue to geophysical exploration and industrial process monitoring.

EIT is modeled using elliptic partial differential equations. Let  $\Omega \subset \mathbb{R}^n$  be a bounded domain representing the body under investigation, and let  $\gamma(x)$  denote its electrical conductivity, assumed to be a positive function. When a voltage  $f$  is applied at the boundary  $\partial\Omega$ , the resulting potential  $u$  inside  $\Omega$  satisfies the conductivity equation

$$\nabla \cdot (\gamma \nabla u) = 0 \quad \text{in } \Omega,$$

with boundary condition  $u|_{\partial\Omega} = f$ . The measurable quantity in practice is the current flux through the boundary, given by  $\gamma \frac{\partial u}{\partial \nu}|_{\partial\Omega}$ , where  $\nu$  denotes the outward unit normal.

This leads naturally to the definition of the Dirichlet-to-Neumann (DtN) map, also called the voltage-to-current map:

$$\Lambda_\gamma : f \mapsto \gamma \frac{\partial u}{\partial \nu} \Big|_{\partial\Omega}.$$

The DtN map encodes all possible boundary measurements and serves as the central object of study in EIT. The inverse problem consists of determining the conductivity  $\gamma(x)$  from knowledge of  $\Lambda_\gamma$ .

This inverse problem is known as the Calderón problem, named after Alberto Calderón, who first formulated it in 1980, one can check the reprint [1]

for historical note. The fundamental question is: does the Dirichlet-to-Neumann map uniquely determine the conductivity inside the domain? In other words, if  $\Lambda_{\gamma_1} = \Lambda_{\gamma_2}$ , does it follow that  $\gamma_1 = \gamma_2$ ?

As physical motivation consider the discrete analogy, suppose  $\partial\Omega$  consists of only  $m$  points. Let  $f_j$  be the voltage at the  $j$ -th boundary point, and let  $\Lambda_\gamma(f)_i$  be the measured current at the  $i$ -th boundary point. The linear map  $f \mapsto \Lambda_\gamma(f)$  takes  $\mathbb{R}^m \rightarrow \mathbb{R}^m$  and can be written as

$$\Lambda_\gamma(f)_i = \sum_{j=1}^m \lambda_{i,j} f_j,$$

where  $\lambda_{i,j}$  is the current at point  $i$  when a unit voltage is applied at point  $j$ . In the continuous setting, the analogous expression is

$$\Lambda_\gamma(f)(x) = \int_{\partial\Omega} \lambda_\gamma(x, y) f(y) dS(y),$$

Knowing  $\Lambda_\gamma(f)$  for all boundary voltages  $f$  is equivalent to knowing the integral kernel  $\lambda_\gamma(x, y)$  for all  $x, y \in \partial\Omega$ . This kernel depends on  $2n - 2$  variables, since  $x$  and  $y$  lie on an  $(n - 1)$ -dimensional manifold.

Thus, in the Calderón inverse problem we aim to recover a function of  $n$  variables (the conductivity  $\gamma$ ) from a function of  $2n - 2$  variables (the kernel  $\lambda_\gamma$ ).

- For  $n = 1$ : The problem is underdetermined ( $\gamma$  has more variables than  $\lambda_\gamma$ ), and indeed we know the DtN map cannot determine the conductivity in one dimension.
- For  $n = 2$ : The problem is well-determined ( $\gamma$  and  $\lambda_\gamma$  both depend on two variables).
- For  $n \geq 3$ : The problem is overdetermined (the data have more degrees of freedom than the unknown).

The Calderón problem has become a central topic in the field of inverse problems and has led to deep developments in analysis, partial differential equations, and geometry. Early results established uniqueness under strong smoothness assumptions on the conductivity. Subsequent work has progressively weakened these assumptions, leading to uniqueness results for less regular conductivities. A key breakthrough was the introduction of complex geometrical optics (CGO) solutions, which allow one to probe the interior of the domain using special solutions to the conductivity equation.

Despite these advances, the problem remains highly non-trivial. In higher dimensions, uniqueness is known under fairly general conditions, but stability,

the sensitivity of the reconstruction to measurement errors is severely ill-posed. The present work, deals with one of the problems regarding stability.

A striking illustration of the severe instability inherent in the Calderón problem is provided by a construction of Bourgain and Wolf in the 1990's, [2]. They developed even further, a result presented by Wolf on [3].

Exhibiting a nontrivial harmonic function in the upper half-space, of class  $C^1$  up to the boundary, whose trace and normal derivative vanish simultaneously on a subset of the boundary with positive measure. Although this phenomenon does not contradict uniqueness results for the Calderón problem, which rely on the full knowledge of the Dirichlet-to-Neumann map, it highlights a fundamental limitation of boundary measurements.

Even smooth, nontrivial solutions may be essentially invisible on large portions of the boundary. This type of degeneracy provides strong intuition for the logarithmic instability that characterizes the inverse problem, as small or localized boundary data may correspond to significant variations in the interior conductivity. In particular, it underscores the challenges that arise in partial data problems, where measurements are only available on subsets of the boundary, and where such “invisible” solutions can severely obstruct stable reconstruction.

In this introducing chapter we wish to present a brief state of the art, to situate the reader on the results mentioned above. A fairly understand of Sobolev spaces and elliptic PDE's is required to understand the proof of the results here presented. Although, not important for our main results on following chapters.

## 1.1 The Calderón Problem

The problem is formally stated as this: let  $\Omega \subset \mathbb{R}^n$  be a domain with smooth boundary. Consider the differential operator

$$L_\gamma = \nabla \cdot (\gamma \nabla), \quad 0 < \varepsilon < \gamma \in C^\infty(\overline{\Omega}),$$

and the associated quadratic form  $Q_\gamma$ , defined by

$$Q_\gamma(f) = \inf_{\substack{w|_{\partial\Omega}=f \\ w \in H^1(\Omega)}} \int_{\Omega} \gamma |\nabla w|^2 dx.$$

Let  $u$  be the unique solution to

$$L_\gamma u = 0 \quad \text{in } \Omega, \quad u|_{\partial\Omega} = f.$$

Then one can show that the infimum is attained at  $u$ , and

$$Q_\gamma(f) = \int_{\Omega} \gamma |\nabla u|^2 dx.$$

Both of these facts are well established at Chapter 2 of [4].

If  $f$  represents a voltage potential imposed on the boundary of a body  $\Omega$  with isotropic conductivity  $\gamma$ , then  $Q_\gamma(f)$  measures the electrical power required to maintain the potential  $f$ . Importantly,  $Q_\gamma(f)$  is a quantity that can be determined solely from boundary measurements.

A direct consequence of the definition is that the unique self-adjoint operator associated with the quadratic form  $Q_\gamma$  is precisely the DtN map on  $\partial\Omega$ :

$$f \mapsto \gamma \frac{\partial u}{\partial \nu} \Big|_{\partial\Omega},$$

The Calderón problem is equivalent to asking if all possible boundary measurements are encoded in the Dirichlet-to-Neumann map  $\Lambda_\gamma$ . At 1987 John Sylvester and Gunther Uhlmann were able to prove a uniqueness result.

**Theorem 1.1** ([5] Theorem 0.1). *Let  $\Omega \subset \mathbb{R}^n$  ( $n \geq 3$ ) be a smooth bounded domain. Suppose  $\gamma_0, \gamma_1 \in C^\infty(\overline{\Omega})$  are strictly positive on  $\overline{\Omega}$ , and assume that*

$$Q_{\gamma_0}(f) = Q_{\gamma_1}(f) \quad \text{for all } f \in H^{1/2}(\partial\Omega).$$

*Then*

$$\gamma_0 = \gamma_1 \quad \text{in } \Omega.$$

Although their techniques were unable to provide a conclusion for  $n = 2$ , it was later on resolved:

**Theorem 1.2** ([4] Theorem 1.2). *Let  $\Omega \subset \mathbb{R}^n$  be a bounded  $C^2$  domain. Let  $\gamma_1, \gamma_2$  be positive functions satisfying:*

$$\gamma_1, \gamma_2 \in \begin{cases} C^2(\overline{\Omega}) & \text{if } n \geq 3, \\ C^2(\overline{\Omega}) \cap L^\infty(\Omega) & \text{if } n = 2. \end{cases}$$

*If  $\Lambda_{\gamma_1} = \Lambda_{\gamma_2}$ , then  $\gamma_1 = \gamma_2$  in  $\Omega$ .*

Also as an algorithm method to reconstruct the solution for  $n \geq 3$ :

**Theorem 1.3** ([4] Theorem 4.28). *Let  $\Omega \subset \mathbb{R}^n$  be a bounded domain with smooth boundary, where  $n \geq 3$ . Assume  $\gamma \in C^2(\overline{\Omega})$  is a strictly positive function. Then, from the knowledge of the Dirichlet-to-Neumann map*

$$\Lambda_\gamma : H^{3/2}(\partial\Omega) \longrightarrow H^{1/2}(\partial\Omega),$$

*it is possible to determine  $\gamma$  in  $\Omega$  in a constructive manner.*

However, implementing EIT reconstruction algorithms in practice is highly challenging due to the diffuse nature of the underlying physics and the severe ill-posedness of the inverse problem: even small measurement errors can produce large distortions in the reconstructed conductivity. In the context of the Calderón problem, this instability can be quantified in an optimal way.

## 1.2 Stability and Partial Data

A fundamental ingredient in the study of the Calderón problem is the principle of unique continuation. Roughly speaking, this property asserts that solutions to elliptic equations cannot vanish to high order without being trivial. In the present context, it ensures that the Dirichlet-to-Neumann map  $\Lambda_\gamma$  encodes global information about the conductivity  $\gamma$ . Indeed, uniqueness results for the inverse problem rely on the fact that boundary measurements propagate into the interior through special solutions, such as complex geometrical optics solutions. These constructions exploit strong unique continuation in the interior of the domain.

Results of this type are obtained by an important analytical reduction in the study of the Calderón problem; its connection to the Schrödinger equation. Given a positive conductivity  $\gamma$ , one can perform the change of variables  $v = \gamma^{1/2}u$ , where  $u$  solves the conductivity equation  $\nabla \cdot (\gamma \nabla u) = 0$ . A direct computation shows that  $v$  satisfies an equation of the form

$$(-\Delta + q)v = 0 \quad \text{in } \Omega,$$

where the potential  $q$  is given by  $q = \gamma^{-1/2} \Delta \gamma^{1/2}$ . This transformation allows one to reinterpret the Calderón problem as an inverse boundary value problem for a Schrödinger operator: recovering the potential  $q$  (and hence  $\gamma$ ) from boundary measurements. Results of this type can be found in a more general setting in a work by David Jerison and Carlos E. Kenig [6], and again by Wolf [7]. In our restricted situation it shows as,

**Theorem 1.4** ([4] Theorem 6.19). *Let  $\Omega \subset \mathbb{R}^n$  be a bounded connected open set with smooth boundary, and let  $q \in L^\infty(\Omega)$ . Assume that  $\Gamma$  is a nonempty open subset of  $\partial\Omega$ . If  $u \in H^2(\Omega)$  satisfies*

$$(-\Delta + q)u = 0 \quad \text{in } \Omega$$

and

$$u|_\Gamma = \partial_\nu u|_\Gamma = 0,$$

then  $u = 0$  in  $\Omega$ .

However, while unique continuation provides rigidity, it does not guarantee stability. As mentioned previously, the inverse map  $\Lambda_\gamma \mapsto \gamma$  is severely ill-posed, with at best logarithmic stability under suitable a priori assumptions. This reflects the fact that the propagation of information from the boundary to the interior is extremely unstable: small perturbations in the boundary data may correspond to large variations of the conductivity. In this sense, uniqueness is a qualitative property, whereas stability is quantitative, and significantly more delicate. Formally, we have

**Theorem 1.5** ([4] Theorem 4.1). *Let  $\Omega \subseteq \mathbb{R}^n$  be a bounded open set with smooth boundary, where  $n \geq 3$ , and let  $\gamma_j$ ,  $j = 1, 2$ , be two positive functions in  $H^{s+2}(\Omega)$  with  $s > n/2$ , satisfying*

$$\frac{1}{M} \leq \gamma_j \leq M,$$

$$\|\gamma_j\|_{H^{s+2}(\Omega)} \leq M.$$

*There are constants  $C = C(\Omega, n, M, s) > 0$  and  $\sigma = \sigma(n, s) \in (0, 1)$  such that*

$$\|\gamma_1 - \gamma_2\|_{L^\infty(\Omega)} \leq \omega(\|\Lambda_{\gamma_1} - \Lambda_{\gamma_2}\|_*)$$

*where  $\omega$  is a modulus of continuity satisfying*

$$\omega(t) \leq C|\log t|^{-\sigma}, \quad 0 < t < 1/e.$$

This instability becomes even more pronounced in the case of partial boundary measurements. In many practical situations, one only has access to  $\Lambda_\gamma$  on a subset of the boundary. The corresponding partial data Calderón problem is substantially more difficult, and additional geometric or analytic assumptions are typically required to recover uniqueness.

From an intuitive perspective, one may think of certain regions of the domain as being poorly “illuminated” by the available boundary data, making them harder to reconstruct. For example when considering partial data subsets the requirement to reconstruct the conductivity  $\gamma$ , or the potential  $q$ , become more restricted, and require methods as Reflection, or Calderon estimates. There are many results and we suggest a look at Chapter 6 by [4], but only to illustrate we suggest two central ones

**Theorem 1.6** ([4] Theorem 6.3). *Let  $\Omega \subset \{x_n > 0\}$  be a bounded open set with smooth boundary, let  $\Gamma_0 = \partial\Omega \cap \{x_n = 0\}$ , and let*

$$\Gamma = \partial\Omega \setminus \Gamma_0.$$

*Let  $q_1, q_2 \in L^\infty(\Omega)$  be such that 0 is not a Dirichlet eigenvalue for  $-\Delta + q_j$  in  $\Omega$ . If one has*

$$\Lambda_{q_1} f|_\Gamma = \Lambda_{q_2} f|_\Gamma \quad \text{for all } f \in H^{1/2}(\partial\Omega) \text{ with } \text{supp}(f) \subset \Gamma,$$

*then  $q_1 = q_2$  in  $\Omega$ .*

**Theorem 1.7** ([4] Theorem 6.15). *Let  $q_1$  and  $q_2$  be two functions in  $L^\infty(\Omega)$  such that the Dirichlet problems for  $-\Delta + q_1$  and  $-\Delta + q_2$  are well-posed. If  $\alpha$  is a unit vector in  $\mathbb{R}^n$  and if*

$$\Lambda_{q_1} f|_{\partial\Omega_{-\varepsilon}} = \Lambda_{q_2} f|_{\partial\Omega_{-\varepsilon}}, \quad \text{for all } f \in H^{1/2}(\partial\Omega)$$

*then  $q_1 = q_2$  in  $\Omega$ .*

It is in this context that the phenomenon exhibited by Bourgain and Wolf [2] becomes particularly relevant. Their construction of a nontrivial harmonic function in the upper half-space, smooth up to the boundary and vanishing together with its normal derivative on a set of positive boundary measure, illustrates a form of boundary invisibility. Although this does not contradict unique continuation in the interior, it shows that boundary information can be highly degenerate on large sets. In particular, such examples provide a compelling heuristic explanation for the instability of the inverse problem: significant interior behavior may produce negligible or even vanishing boundary signatures on substantial portions of the boundary.

A useful physical interpretation of Bourgain's and Wolf example further highlights the instability of the inverse problem, even in the simplest setting of a homogeneous medium. In this case, the conductivity is constant,  $\gamma \equiv 1$ , and the boundary flux reduces to the normal derivative  $\partial_\nu u$ . Their construction shows that there exist nontrivial harmonic functions whose trace and normal derivative vanish simultaneously on a subset of the boundary of positive measure. From a physical perspective, this means that on a substantial portion of the boundary there is neither applied voltage nor resulting current, yet the potential inside the domain is nonzero. In other words, part of the interior becomes completely undetectable from these boundary measurements.

It is worth emphasizing that the phenomenon exhibited by Bourgain is inherently dimension-dependent. In dimension  $n \geq 3$ , one can construct nontrivial harmonic functions whose trace and normal derivative vanish simultaneously on a subset of the boundary of positive measure. In contrast, such behavior is impossible in dimension  $n = 2$ : if both the Dirichlet and Neumann data of a harmonic function vanish on a set of positive boundary measure, then the solution must be identically zero. This reflects a stronger form of boundary unique continuation in two dimensions, and is consistent with the enhanced rigidity observed in the two-dimensional Calderón problem, where uniqueness holds under significantly weaker regularity assumptions; one can check Chapter 5 of [4].

### 1.3 The case of Lipschitz Domains

The work of Bourgain and Wolff was not originally motivated by the Calderón problem, but rather by fundamental questions in harmonic analysis concerning boundary behavior and uniqueness for harmonic functions. More precisely, they investigated to what extent boundary data determines a harmonic function, and whether it is possible for a nontrivial solution to exhibit strong vanishing properties on large subsets of the boundary. Their construction shows that, in dimensions  $n \geq 3$ , there exist nontrivial harmonic functions in the upper half-space that are  $C^1$  up to the boundary and whose trace and normal derivative vanish simultaneously on a set of positive boundary measure. This result demonstrates a

surprising failure of boundary determination in a measure-theoretic sense, revealing that large portions of the boundary may carry no information about certain interior configurations.

The restriction to the upper half-space  $\mathcal{H}^d = \{x_d > 0\}$  is both natural and technically advantageous. From a conceptual standpoint, the half-space provides a canonical local model for smooth domains near the boundary, allowing one to isolate and analyze boundary phenomena without geometric complications. Moreover, its flat boundary enables the use of powerful tools from harmonic analysis, such as Fourier transform methods and explicit representations of harmonic functions via the Poisson kernel. This setting also allows for precise control of boundary regularity and facilitates the construction of counterexamples with prescribed vanishing properties.

Nevertheless, Wenshang Wang was able to extend their result for  $C^{1,s}$  domains throughout Kelvin transformations at [8]. Moreover, on a work by L. Caffarelli and A. Friedman [9] they were able to estimate the Hausdorff dimension of the critical set on the interior to be less or equal than  $n - 2$ .

CHAPTER 2

# The case $\mathcal{H}^2$

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In this chapter, we begin by introducing some concepts and notation, which will be used through the text. We observe that in this chapter  $H^p(dt)$  denotes the Hardy space which will be defined later on. Rather than the fractional Sobolev space from the previous chapter. The final result will come as consequence of the following theorem:

**Theorem 2.1.** *Let  $h(t)$  be a nonnegative function on  $L^p(\mathbb{R})$ . Then there is  $f(z) \in H^p(dt)$  such that  $|f(t)| = h(t)$  almost everywhere if and only if*

$$\int \frac{\log h(t)}{1+t^2} dt > -\infty. \quad (2.1)$$

The first two parts of this chapter are dedicated to introduce the tools necessary to understand and prove Theorem 2.1.

## 2.1 Preliminaries

Many of the results that will be presented on this section are elementary, in a sense that it can be easily found on your favorite complex analysis book. We will present them to achieve the goal of proving the final main result, in this sense they are here serving as tools for an end. Except when is possible to provide some details on the arguments, that are not present on the references, or when needed to do a simplified, or adapted version of the results, we are going to reference their proofs to avoid any pedantry.

### 2.1.1 Poisson Kernels

Poisson kernels will be a major tool along this text, we are going to present a few major results that will be used in the following of this chapter. Although we focus solely on the upper half plane, many results are easier to state on the disc.

Therefore considering the conformal maps:

$$\Phi : D \rightarrow \mathcal{H}^2, \quad \Phi(\zeta) = i \frac{1 + \zeta}{1 - \zeta}, \quad \Phi^{-1}(z) = \frac{z - i}{z + i},$$

noting as well, their behavior on the boundary,

$$\Phi(e^{i\theta}) = \tan\left(\frac{\theta}{2}\right), \quad e^{i\theta} = \frac{t - i}{t + i}, \quad t \in \mathbb{R},$$

and the useful change of variables formula, between the disk and the upper-half plane

$$\frac{d\theta}{2\pi} = \frac{1}{\pi} \frac{dt}{1 + t^2}.$$

we are going to introduce Poisson kernels in both approaches, as convenient.

The Poisson kernel arises as the mechanism that reconstructs a harmonic function inside a domain from its boundary values. Its defining feature is that it assigns to each interior point a probability measure on the boundary, describing how strongly each boundary point influences the value of the function at that interior point.

Let  $u$  be harmonic in the unit disc  $D$ . For  $z \in D$ , the value  $u(z)$  is a weighted average of the boundary values of  $u$ ,

$$u(z) = \int_0^{2\pi} u(e^{i\theta}) P_z(\theta) \frac{d\theta}{2\pi}, \quad P_z(\theta) = \frac{1 - |z|^2}{|e^{i\theta} - z|^2}.$$

The kernel  $P_z(\theta)$  is positive, integrates to one, and becomes increasingly concentrated near the boundary point closest to  $z$  as  $|z| \rightarrow 1$ . Thus boundary points near the radial projection of  $z$  contribute the most to the value of  $u(z)$ .

An analogous phenomenon occurs in the upper half-plane  $\mathcal{H}^2$ . If  $u$  is harmonic in it, then

$$u(x + iy) = \int_{\mathbb{R}} u(t) P_y(x - t) dt, \quad P_y(x - t) = \frac{1}{\pi} \frac{y}{(x - t)^2 + y^2}.$$

The formula for this kernel can be obtained through the conformal mappings introduced above.

Here the Poisson kernel again defines a probability measure on the boundary  $\mathbb{R}$ . The parameter  $y$  measures the vertical distance to the boundary, and as  $y \rightarrow 0^+$  the kernel concentrates near the point  $t = x$ . Thus the value of  $u(x + iy)$  is determined primarily by boundary values close to  $x$ .

Associated to the Poisson kernel on the upper half-plane,

$$P_y(t) = \frac{1}{\pi} \frac{y}{t^2 + y^2},$$

there is a naturally defined *conjugate* (or *dual*) kernel, called the *conjugate Poisson kernel*,

$$Q_y(t) = \frac{1}{\pi} \frac{t}{t^2 + y^2}.$$

In contrast with  $P_y$ , the kernel  $Q_y$  is odd and satisfies

$$\int_{\mathbb{R}} Q_y(t) dt = 0.$$

It does not define a probability measure; instead, it describes the signed influence of boundary values and is responsible for the construction of harmonic conjugates.

Let  $u$  be a harmonic function on  $\mathcal{H}^2$  whose boundary values belong to  $L^p(\mathbb{R})$  for some  $1 < p < \infty$ . Define

$$v(x + iy) = \int_{\mathbb{R}} u(t) Q_y(x - t) dt.$$

Then  $v$  is harmonic in  $\mathcal{H}^2$ , and the function

$$F(z) = u(z) + i v(z)$$

is holomorphic in the upper half-plane. In particular,  $v$  is a harmonic conjugate of  $u$ , uniquely determined up to an additive constant.

The following theorems provide more understanding and formalize this discussion. They also will be very helpful to prove results later on.

**Theorem 2.2** ([10] Theorem I.3.1). *Let  $P_y$  be the Poisson kernel on  $\mathbb{R}$ .*

1. *If  $1 \leq p < \infty$  and  $f \in L^p(\mathbb{R})$ , then*

$$\|P_y * f - f\|_{L^p} \longrightarrow 0 \quad (y \rightarrow 0).$$

2. *If  $f$  is bounded and uniformly continuous on  $\mathbb{R}$ , then  $P_y * f$  converges uniformly to  $f$ .*

*Remark 2.3.* An important observation here is that, since our primary goal is to prove a result in  $L^2$  (check 2.32), the theorem above as many other later on, have appropriate statements to treat the cases  $p = \infty$ , which we ignore.

**Theorem 2.4** ([10] Theorem I.3.5). *Let  $u(z)$  be a harmonic function on the upper half plane  $\mathcal{H}^2$ . If  $1 < p \leq \infty$ ,  $u$  is the Poisson integral of a function in  $L^p$  if and only if*

$$\sup_y \|u(x, y)\|_{L^p(dx)} < \infty.$$

### 2.1.2 Nontangential Limits

The proof of Theorem 2.1 lies on the construction of several results, on which, subharmonic functions and the existence of nontangential limits plays a key role. For a fixed  $\alpha > 0$  we consider cones,  $\Gamma_\alpha(t) := \{z \in \mathcal{H}^2 : |x - t| < \alpha y\}$  for  $z = x + iy$  and  $t \in \mathbb{R}$ . We then define:

**Definition 2.5.** If  $u(z)$  is harmonic on  $\mathcal{H}^2$ , we set

$$u^*(t) = \sup_{z \in \Gamma_\alpha(t)} |u(z)|,$$

as the nontangential maximal function of  $u$  at  $t \in \mathbb{R}$ .

At first, this definition depends on the choice of  $\alpha$ , but since it has been fixed it, we ignore that distinction.

We also introduce the well known Hardy-Littlewood maximal operator, which plays a minor role on this work. Nevertheless, it will still be useful for some inequalities in this preliminary moment.

**Definition 2.6.** For  $f \in L^1_{loc}$  we define

$$Mf(x) = \sup_{x \in U} \frac{1}{|U|} \int_U |f(y)| dy$$

as the Hardy Littlewood maximal function, where the supremum is taken over open sets that contains  $x$ .

Some very basic results about this operator are:

**Theorem 2.7** ([11] Chapter 3 - Theorem 1.1). *Suppose  $f$  integrable in  $\mathbb{R}^d$ . Then:*

1.  $Mf$  is measurable.
2.  $Mf(x) < \infty$  for a.e  $x$ .
3.  $|\{x \in \mathbb{R}^d : Mf(x) > \alpha\}| \leq \frac{A}{\alpha} \|f\|_1$ .

Many important, and standard results can be stated about Hardy-Littlewood maximal operator; for the sake of our objectives we recommend any unfamiliar reader to take a quick look at [10],[11]. Regarding everything that has been introduced so far, we have:

**Theorem 2.8.** *Let  $f \in L^1(dt/(1+t^2))$  and let  $u(x, y)$  be the Poisson integral of  $f(t)$ ,*

$$u(x, y) = \int P_y(s) f(x - s) ds.$$

Then

$$u^*(t) = \sup_{\Gamma_\alpha(t)} |u(x, y)| \leq A_\alpha M f(t), \quad t \in \mathbb{R},$$

where  $A_\alpha$  is a constant depending only on  $\alpha$ .

*Proof.* Given  $t \in \mathbb{R}$ , define the shifted function

$$f_t(s) = f(s + t), \quad s \in \mathbb{R}.$$

Let  $u_{f_t}$  denote the Poisson integral of  $f_t$ :

$$u_{f_t}(x, y) = \int_{\mathbb{R}} P_y(s) f_t(x - s) ds.$$

Now note the translation property of the cone:

$$(x, y) \in \Gamma_\alpha(t) \quad \Leftrightarrow \quad (x - t, y) \in \Gamma_\alpha(0).$$

For  $(x, y) \in \Gamma_\alpha(t)$ , let  $\tilde{x} = x - t \in \Gamma_\alpha(0)$ . Then

$$\begin{aligned} u(x, y) &= \int_{\mathbb{R}} P_y(s) f(x - s) ds \\ &= \int_{\mathbb{R}} P_y(s) f(\tilde{x} + t - s) ds \\ &= \int_{\mathbb{R}} P_y(s) f_t(\tilde{x} - s) ds \\ &= u_{f_t}(\tilde{x}, y). \end{aligned}$$

Thus

$$\sup_{(x, y) \in \Gamma_\alpha(t)} |u(x, y)| = \sup_{(\tilde{x}, y) \in \Gamma_\alpha(0)} |u_{f_t}(\tilde{x}, y)|.$$

For the maximal function, we have

$$M f(t) = \sup_{h>0} \frac{1}{2h} \int_{t-h}^{t+h} |f(s)| ds = \sup_{h>0} \frac{1}{2h} \int_{-h}^h |f(w + t)| dw = M f_t(0).$$

If the theorem holds for  $t = 0$ , then applying it to the function  $f_t$  gives

$$\sup_{\Gamma_\alpha(0)} |u_{f_t}(x, y)| \leq A_\alpha M f_t(0).$$

Therefore,

$$\sup_{\Gamma_\alpha(t)} |u(x, y)| \leq A_\alpha M f_t(0) = A_\alpha M f(t).$$

Hence it suffices to prove the theorem for the case  $t = 0$ . We want to control  $|u(x, y)|$  for  $(x, y) \in \Gamma_\alpha(0)$ , i.e.,  $|x| < \alpha y$ . Note that

$$|u(x, y)| \leq \int_{\mathbb{R}} P_y(x - s) |f(s)| ds.$$

If we can dominate  $P_y(x - s)$  by an even, positive function  $\psi(s)$  decreasing on  $s > 0$  that is a convex combination of box kernels, then the integral will be bounded by  $(\int \psi) \cdot Mf(0)$ .

For the axis case  $x = 0$  we have  $u(0, y) = \int P_y(s)f(s) ds$  with  $P_y(s)$  even, positive, and decreasing in  $|s|$ . Such a function can be approximated from below by step functions

$$h_n(s) = \sum_{j=1}^N a_j \chi_{(-x_j, x_j)}(s), \quad a_j \geq 0,$$

with  $\sum_j 2x_j a_j \leq 1$  (since  $\int P_y = 1$ ). Then

$$\begin{aligned} \int h_n(s)|f(s)| ds &= \sum a_j(2x_j) \cdot \frac{1}{2x_j} \int_{-x_j}^{x_j} |f(s)| ds \\ &\leq \left( \sum 2x_j a_j \right) Mf(0) \\ &\leq Mf(0). \end{aligned}$$

Monotone convergence gives  $|u(0, y)| \leq Mf(0)$ . Now, for off-axis points, that is  $(x, y)$  with  $|x| < \alpha y$ , we need a symmetric majorant for  $P_y(x - s)$ . Define

$$\psi(s) = \sup_{|t| \geq |s|} P_y(x - t), \quad s \in \mathbb{R}.$$

Then  $\psi$  is even, positive, and decreasing for  $s > 0$ , and  $P_y(x - s) \leq \psi(s)$  for all  $s$ . Assume without loss  $x \geq 0$  (by symmetry). For  $s \geq 0$ :

- If  $s \leq x$ , the point  $t = x$  satisfies  $|t| \geq s$ , giving  $P_y(0) = \frac{1}{\pi y}$  as the supremum.
- If  $s > x$ , the closest point to  $x$  with  $|t| \geq s$  is  $t = s$  (since  $x \geq 0$ ), giving minimal distance  $s - x$ .

Thus

$$\psi(s) = \begin{cases} \frac{1}{\pi y}, & 0 \leq s \leq x, \\ \frac{y}{\pi[(s - x)^2 + y^2]}, & s > x. \end{cases}$$

Since  $\psi$  is even,

$$\int_{\mathbb{R}} \psi(s) ds = 2 \int_0^\infty \psi(s) ds.$$

Compute:

$$\int_0^\infty \psi(s) ds = \int_0^x \frac{1}{\pi y} ds + \int_x^\infty \frac{y}{\pi[(s - x)^2 + y^2]} ds = \frac{x}{\pi y} + \frac{1}{2}.$$

Therefore,

$$\int_{\mathbb{R}} \psi(s) ds = \frac{2x}{\pi y} + 1.$$

Since  $|x| < \alpha y$ , we have

$$\int_{\mathbb{R}} \psi(s) ds \leq 1 + \frac{2\alpha}{\pi} = A_\alpha.$$

Approximate  $\psi$  from below by step functions  $h_n$  as before, each a convex combination of box kernels with total mass  $\leq A_\alpha$ . Then

$$\int \psi(s) |f(s)| ds \leq A_\alpha \cdot Mf(0).$$

Hence

$$|u(x, y)| \leq \int \psi(s) |f(s)| ds \leq A_\alpha Mf(0).$$

This holds for all  $(x, y) \in \Gamma_\alpha(0)$ , so

$$\sup_{\Gamma_\alpha(0)} |u(x, y)| \leq A_\alpha Mf(0).$$

Translating back to general  $t$  completes the proof.  $\square$

We decided to present a proof for the theorem above, to give some more detailing. The main theorem that we are interested in is about the existence almost everywhere of nontangential limits of the form  $\lim_{\Gamma_\alpha(t) \ni z \rightarrow t} u(z)$  (check again the statement of 2.32). First we are going to need:

**Theorem 2.9.** *Let  $u(x)$  be harmonic on  $\mathcal{H}^2$  and let  $1 < p < \infty$ . Assume*

$$\sup_y \int |u(x + iy)|^p dx < \infty.$$

*Then  $u^*(t) \in L^p$ , and*

$$\|u^*\|_p^p \leq B_{p,\alpha} \sup_y \int |u(x + iy)|^p dx. \quad (2.2)$$

*The constants  $B_{p,\alpha}$  depend only on  $p$  and  $\alpha$ .*

*Proof.* In view of Theorem 2.4,  $u(z)$  is the Poisson integral for a function  $f(t) \in L^1(\mathbb{R})$ ,

$$u(z) = \int P_y(x - t) f(t) dt.$$

Moreover, by an approximation argument, theorem 2.2 we have that  $\|P_y * f - f\|_p \rightarrow 0$  as  $y \rightarrow 0$ . Therefore

$$\|f\|_p = \lim_{y \rightarrow 0} \|P_y * f\|_p = \lim_{y \rightarrow 0} \|u(\cdot + iy)\|_p \leq \sup_y \|u(\cdot + iy)\|_p.$$

By Theorem 2.8 and the strong type inequality for Hardy-Littlewood maximal operator (check Remark ??), we have  $\|u^*\|_p \leq A_\alpha \|Mf\|_p \leq B_{p,\alpha} \|f\|_p$ . Putting everything together

$$\|u^*\|_p^p \leq B_{p,\alpha} \sup_y \int |u(x + iy)|^p dx.$$

□

*Remark 2.10.* On the previous proof, when invoking Theorem 2.8 one might notice that we have  $f \in L^{p>1}(\mathbb{R})$ , instead of  $L^1(dt/(1+t^2))$  as stated there. However, it actually holds that  $L^{p>1}(\mathbb{R}) \hookrightarrow L^1(dt/(1+t^2))$ . Indeed,

$$\int \frac{|f(t)|}{1+t^2} dt \leq \|f\|_p \left\| \frac{1}{1+t^2} \right\|_q,$$

where  $q = \frac{p}{p-1}$ . But  $\left\| \frac{1}{1+t^2} \right\|_q < \infty \Leftrightarrow q > \frac{1}{2}$ , which is always true.

**Theorem 2.11** (Fatou). *Let  $u(z)$  be harmonic on  $\mathcal{H}^2$  and  $1 < p < \infty$ . Assume that  $\sup_y \|u(x + iy)\|_{L^p(dx)} < \infty$ . Then for almost  $t \in \mathbb{R}$  there exists a nontangential limit*

$$\lim_{\Gamma_\alpha(t) \ni z \rightarrow t} u(z) = f(t).$$

*Moreover,  $u(z)$  is the Poisson integral of the boundary value function  $f(t)$ ; we also have  $\|u(x + iy) - f(x)\|_p \rightarrow 0$  as  $y \rightarrow 0$ .*

*Proof.* Again, in view of Theorem 2.4,  $u(z)$  is the Poisson integral of a function  $f(t) \in L^p$ . We will show  $u(z)$  has nontangential limit  $f(t)$  for almost all  $t$ . We can assume  $f$  is real valued. Let

$$\Omega_f(t) = \limsup_{z \rightarrow t} u(z) - \liminf_{z \rightarrow t} u(z),$$

where  $z$  is constrained to  $\Gamma_\alpha(t)$ . Then by Theorem 2.9  $\Omega_f(t) \leq 2u^*(t) \leq 2A_\alpha Mf(t)$ , so that  $\Omega_f$ , as well as the limes superior and the limes inferior, is finite almost everywhere. The function  $\Omega_f(t)$  represents the nontangential oscillation of  $u$  at  $t$ , and  $u$  has a nontangential limit at  $t$  if and only if  $\Omega_f(t) = 0$ .

By Theorem 2.9, and by Chebychev's inequality if  $p > 1$ , we have

$$|\{t : \Omega_f(t) > \varepsilon\}| \leq B_p \left( \frac{2}{\varepsilon} \|f\|_{L^p} \right)^p.$$

Now if  $g \in L^p$  and if in addition  $g \in C_c(\mathbb{R})$ , then by Theorem 2.2,  $\Omega_g = 0$  for all  $t$ , and so  $\Omega_f = \Omega_{f+g}$  (linearity of the convolution and the integral). Take  $g \in C_c(\mathbb{R})$  so that  $\|f + g\|_{L^p} \leq \varepsilon^2$ . Then

$$|\{t : \Omega_f(t) > \varepsilon\}| = |\{t : \Omega_{f+g}(t) > \varepsilon\}| \leq B_p \left( \frac{2}{\varepsilon} \|f + g\|_{L^p} \right)^p \leq c_p \varepsilon^p.$$

Consequently  $\Omega_f(t) = 0$  almost everywhere and  $u$  has a nontangential limit almost everywhere. The limit coincides with  $f(t)$  almost everywhere because  $u(x, y)$  converges in  $L^p$  norm to  $f(x)$ . That proves the theorem.  $\square$

### 2.1.3 Subharmonic Functions

We still have some remarks to make before concluding the technicalities. Define and state some useful results about subharmonic functions are the last on that list. We wish to argue about subharmonicity of  $|f|^p, \log(|f|)$  in the case where  $f$  is holomorphic, along with some technical lemma.

**Definition 2.12.** Let  $V$  be an open set in the plane. A **subharmonic function** on  $V$  is a function  $v : V \rightarrow [-\infty, \infty)$  such that

1.  $v$  is upper semicontinuous:

$$v(z_0) \geq \limsup_{z \rightarrow z_0} v(z), \quad z_0 \in V.$$

2. For each  $z_0 \in V$  there is  $r_{z_0} > 0$  such that the disc  $D(z_0, r_{z_0}) = \{z : |z - z_0| < r_{z_0}\}$  is contained in  $V$  and such that for every  $r < r_{z_0}$ ,

$$v(z_0) \leq \frac{1}{\pi r^2} \int_{|z - z_0| < r} v(z) dz$$

**Theorem 2.13** ([10] Theorem 6.2). *Let  $v(z)$  be a subharmonic function on  $\Omega$ , and let  $\varphi(t)$  be an increasing convex function on  $[-\infty, \infty)$ , continuous at  $t = -\infty$ . Then  $\varphi \circ v$  is a subharmonic function on  $\Omega$ .*

As a consequence of the previous theorem, we have

**Theorem 2.14.** *Let  $\Omega \subset \mathbb{C}$  be a domain and let  $f$  be analytic on  $\Omega$ . Then:*

1.  $\log |f|$  is subharmonic on  $\Omega$ ;
2. for every  $p \geq 1$ , the function  $|f|^p$  is subharmonic on  $\Omega$ .

*Proof.* We begin with the logarithmic case. Define

$$v(z) := \log |f(z)|,$$

with the convention  $v(z) = -\infty$  whenever  $f(z) = 0$ .

Since  $f$  is continuous,  $|f|$  is continuous and  $\log |f|$  is continuous on the set  $\{z \in \Omega : f(z) \neq 0\}$ . If  $z_0$  is a zero of  $f$ , then  $|f(z)| \rightarrow 0$  as  $z \rightarrow z_0$ , hence

$$\lim_{z \rightarrow z_0} \log |f(z)| = -\infty = v(z_0),$$

which implies

$$v(z_0) \geq \limsup_{z \rightarrow z_0} v(z).$$

Therefore  $v$  is upper semicontinuous on  $\Omega$ . Let  $z_0 \in \Omega$  be such that  $f(z_0) \neq 0$ .

Since zeros of analytic functions are isolated, there exists  $r > 0$  such that

$$f(z) \neq 0 \quad \text{for all } |z - z_0| < r.$$

On the disk  $D(z_0, r)$ , the function  $\log |f|$  is harmonic, since locally  $f = e^g$  with  $g$  holomorphic and  $\log |f| = \operatorname{Re}(g)$ . Hence the mean value property gives

$$\log |f(z_0)| = \frac{1}{\pi r^2} \iint_{|z - z_0| < r} \log |f(z)| \, dx \, dy.$$

In particular, the sub-mean inequality holds. Let  $z_0 \in \Omega$  with  $f(z_0) = 0$ . Then  $v(z_0) = -\infty$ .

For any  $r > 0$  such that  $D(z_0, r) \subset \Omega$ , we have

$$v(z_0) = -\infty \leq \frac{1}{\pi r^2} \iint_{|z - z_0| < r} \log |f(z)| \, dx \, dy,$$

provided the integral is finite. To verify this, write locally

$$f(z) = (z - z_0)^m g(z),$$

where  $m \geq 1$  and  $g$  is analytic with  $g(z_0) \neq 0$ . Then

$$\log |f(z)| = m \log |z - z_0| + \log |g(z)|.$$

Since  $\log |z| \in L^1_{\text{loc}}(\mathbb{C})$  and  $\log |g|$  is continuous,  $\log |f|$  is locally integrable, and the integral is finite. Thus the sub-mean inequality holds at zeros as well.

The function  $\log |f|$  is upper semicontinuous and satisfies the sub-mean inequality at every point of  $\Omega$ . Hence  $\log |f|$  is subharmonic on  $\Omega$ .

Now, we deal with the case  $|f|^p$ ,  $p \geq 1$ . Set

$$v(z) = \log |f(z)|.$$

We have shown that  $v$  is subharmonic. Define

$$\varphi(t) = e^{pt}, \quad t \in [-\infty, \infty).$$

Then  $\varphi$  is increasing, convex, and continuous at  $t = -\infty$  with  $\varphi(-\infty) = 0$ .

By Theorem 2.14, the composition  $\varphi \circ v$  is subharmonic. Since

$$(\varphi \circ v)(z) = e^{p \log |f(z)|} = |f(z)|^p,$$

it follows that  $|f|^p$  is subharmonic on  $\Omega$ . □

*Remark 2.15.* Note that, since  $\varphi(t) = \max\{t, 0\}$  is convex we also have  $\log^+ |f|$  subharmonic.

**Theorem 2.16** ([10] Theorem I.6.3). *Let  $v : \Omega \rightarrow [-\infty, \infty)$  be an upper semi-continuous function. Then  $v$  is subharmonic on  $\Omega$  if and only if the following condition holds: If  $u(z)$  is a harmonic function on a bounded open subset  $W$  of  $\Omega$  and if*

$$\lim_{W \ni z \rightarrow \zeta} (v(z) - u(z)) \leq 0$$

*for all  $\zeta \in \partial W$ , then*

$$v(z) \leq u(z), \quad z \in W.$$

**Lemma 2.17** (Poisson majorization of a subharmonic function). *Let  $v$  be a subharmonic function in the upper half-plane  $\mathcal{H}^2$ . Fix  $r > 0$ . Then, for every  $z = x + iy \in \mathcal{H}^2$ , one has*

$$v(x + iy) \leq \int_{\mathbb{R}} v(t + ir) P_{y-r}(x - t) dt,$$

where

$$P_y(t) = \frac{1}{\pi} \frac{y}{t^2 + y^2}$$

is the Poisson kernel.

*Proof.* Fix  $r > 0$  and define the translated function

$$w(\zeta) = v(\zeta + ir), \quad \zeta \in \mathcal{H}^2.$$

Since translations preserve subharmonicity,  $w$  is subharmonic in  $\mathcal{H}^2$ .

Now define

$$U(\zeta) = \int_{\mathbb{R}} w(t) P_{\text{Im}(\zeta)}(\text{Re}(\zeta) - t) dt, \quad \zeta \in \mathcal{H}^2.$$

The function  $U$  is harmonic, being the Poisson integral of the boundary data  $w|_{\mathbb{R}}$ .

Moreover, for almost every  $t \in \mathbb{R}$ ,

$$\lim_{\eta \rightarrow 0^+} U(t + i\eta) = w(t),$$

so  $U$  admits  $w$  as its boundary values in the sense of nontangential limits.

Since  $w$  is subharmonic and  $U$  is harmonic with the same boundary values, Theorem 2.16 implies that

$$w(\zeta) \leq U(\zeta), \quad \zeta \in \mathcal{H}^2.$$

Taking  $\zeta = x + i(y - r)$  (with  $y > r$ ), we obtain

$$v(x + iy) = w(x + i(y - r)) \leq \int_{\mathbb{R}} v(t + ir) P_{y-r}(x - t) dt.$$

□

**Lemma 2.18** (Subharmonic Poisson inequality via boundary  $\liminf$ ). *Let  $v$  be a subharmonic function in the upper half-plane  $\mathcal{H}^2$ . Assume that*

$$\sup_{y>0} \int_{\mathbb{R}} v^+(x + iy) dx < \infty.$$

*Then, for every  $z = x + iy \in \mathcal{H}^2$ , one has*

$$v(x + iy) \leq \liminf_{r \rightarrow 0^+} \int_{\mathbb{R}} v(t + ir) P_y(x - t) dt,$$

*where*

$$P_y(t) = \frac{1}{\pi} \frac{y}{t^2 + y^2}$$

*is the Poisson kernel of the upper half-plane.*

*Proof.* Fix a point  $z_0 = x + iy \in \mathcal{H}^2$  with  $y > 0$ . Let  $r$  satisfy  $0 < r < y$ . By Lemma 2.17, we have

$$v(x + iy) \leq \int_{\mathbb{R}} v(t + ir) P_{y-r}(x - t) dt. \quad (1)$$

Define

$$I_r := \int_{\mathbb{R}} v(t + ir) P_{y-r}(x - t) dt, \quad 0 < r < y.$$

Then (1) can be written as

$$v(x + iy) \leq I_r, \quad 0 < r < y. \quad (2)$$

For each fixed  $t \in \mathbb{R}$ ,

$$\lim_{r \rightarrow 0^+} P_{y-r}(x - t) = P_y(x - t),$$

since

$$P_{y-r}(s) = \frac{1}{\pi} \frac{y - r}{s^2 + (y - r)^2}$$

depends continuously on  $r$  for  $r < y$ . Fix  $y > 0$ . For all  $0 < r < y/2$  and all  $t \in \mathbb{R}$ , one has

$$P_{y-r}(x - t) \leq C_y P_y(x - t),$$

where one may take  $C_y = 4$ . Indeed,

$$\frac{y - r}{(x - t)^2 + (y - r)^2} \leq \frac{y}{(x - t)^2 + y^2/4} \leq 4 \frac{y}{(x - t)^2 + y^2}.$$

Write

$$v(t + ir) = v^+(t + ir) - v^-(t + ir).$$

Using the domination above, we obtain

$$v^+(t + ir) P_{y-r}(x - t) \leq C_y v^+(t + ir) P_y(x - t).$$

Since  $P_y$  is nonnegative and satisfies

$$\int_{\mathbb{R}} P_y(x - t) dt = 1,$$

it follows that

$$\int_{\mathbb{R}} v^+(t + ir) P_y(x - t) dt \leq \int_{\mathbb{R}} v^+(t + ir) dt \leq \sup_{s > 0} \int_{\mathbb{R}} v^+(t + is) dt < \infty.$$

Thus the family

$$\{v^+(t + ir) P_{y-r}(x - t)\}_{0 < r < y/2}$$

is uniformly integrable in  $t$ . Using this together with Fatou's lemma on the negative part gives

$$\liminf_{r \rightarrow 0^+} I_r \geq \int_{\mathbb{R}} \liminf_{r \rightarrow 0^+} v(t + ir) P_y(x - t) dt. \quad (3)$$

From (2), we have  $v(x + iy) \leq I_r$  for all  $0 < r < y$ . Taking the  $\liminf$  as  $r \rightarrow 0^+$  and using (3), we obtain

$$v(x + iy) \leq \liminf_{r \rightarrow 0^+} \int_{\mathbb{R}} v(t + ir) P_y(x - t) dt.$$

This proves the lemma.  $\square$

## 2.2 The Hardy Space

Hardy spaces provide a natural framework for the study of boundary values of holomorphic functions, harmonic conjugates, and singular integral operators. For  $0 < p < \infty$ , the Hardy space  $H^p(\mathcal{H}^2) = H^p(dt)$  is defined as the space of holomorphic functions  $F : \mathcal{H}^2 \rightarrow \mathbb{C}$  such that

$$\|F\|_{H^p(dt)} := \sup_{y>0} \left( \int_{\mathbb{R}} |F(x + iy)|^p dx \right)^{1/p} < \infty. \quad (2.3)$$

For  $p = \infty$ ,  $H^\infty(dt)$  consists of bounded holomorphic functions on  $\mathcal{H}^2$ .

A fundamental feature of Hardy spaces is that the defining growth condition controls the boundary behavior of the function. If  $F \in H^p(dt)$  with  $0 < p < \infty$ , then the nontangential limits

$$F(x) := \lim_{\substack{z \rightarrow x \\ z \in \Gamma(x)}} F(z)$$

exist for almost every  $x \in \mathbb{R}$ , where  $\Gamma(x)$  denotes a nontangential approach region at  $x$ . Moreover, the boundary function  $F|_{\mathbb{R}}$  belongs to  $L^p(\mathbb{R})$ , and

$$\|F\|_{H^p(dt)} = \|F|_{\mathbb{R}}\|_{L^p(\mathbb{R})}.$$

Thus the Hardy space norm depends only on the boundary values of the function.

### 2.2.1 Boundary Behavior.

As mentioned previously, the  $H^p$  norm of a Hardy function is determined by the  $L^p$  norm of its boundary values. Formally we have

**Theorem 2.19.** *Let  $1 < p < \infty$  and let  $f(z)$  be a function in  $H^p(dt)$ . Then for any  $\alpha > 0$ , the nontangential maximal function*

$$f^*(t) = \sup_{z \in \Gamma_\alpha(t)} |f(z)|$$

*is in  $L^p(\mathbb{R})$  and*

$$\|f^*\|_p^p \leq A_\alpha \|f\|_{H^p}^p,$$

*where the constant  $A_\alpha$  depends on  $\alpha$  and  $p$ . Moreover, for almost all  $t \in \mathbb{R}$ ,  $f(z)$  has a nontangential limit  $f(t) \in L^p(\mathbb{R})$  satisfying*

$$\int |f(t)|^p dt = \|f\|_p^p = \|f\|_{H^p}^p$$

and

$$\lim_{y \rightarrow 0} \|f(t + iy) - f(t)\|_p^p = 0.$$

*Proof.* Write  $f(z) = u(z) + iv(z)$  where  $u, v$  are harmonic on  $\mathcal{H}^2$ . For any  $y > 0$ , we have

$$|u(x + iy)| \leq |f(x + iy)|, \quad |v(x + iy)| \leq |f(x + iy)|.$$

Thus,

$$\sup_{y>0} \int_{\mathbb{R}} |u(x + iy)|^p dx \leq \|f\|_{H^p}^p,$$

and the same holds for  $v$ .

By Theorem 2.9 for harmonic functions, the nontangential maximal functions

$$u^*(t) = \sup_{z \in \Gamma_\alpha(t)} |u(z)|, \quad v^*(t) = \sup_{z \in \Gamma_\alpha(t)} |v(z)|$$

satisfy

$$\|u^*\|_p^p \leq B_{p,\alpha} \sup_{y>0} \|u(\cdot + iy)\|_p^p \leq B_{p,\alpha} \|f\|_{H^p}^p,$$

and similarly

$$\|v^*\|_p^p \leq B_{p,\alpha} \|f\|_{H^p}^p.$$

Since  $|f(z)| \leq |u(z)| + |v(z)|$ , we have pointwise

$$f^*(t) \leq u^*(t) + v^*(t).$$

By Minkowski's inequality,

$$\|f^*\|_p \leq \|u^*\|_p + \|v^*\|_p.$$

Putting all together,

$$\|f^*\|_p^p \leq 2^{p-1} (\|u^*\|_p^p + \|v^*\|_p^p) \leq 2^p B_{p,\alpha} \|f\|_{H^p}^p.$$

Thus, taking  $A_{p,\alpha} = 2^p B_{p,\alpha}$ , we obtain

$$\|f^*\|_p^p \leq A_{p,\alpha} \|f\|_{H^p}^p.$$

This proves the maximal inequality. By Theorem 2.11, the boundary function  $f \in L^p(\mathbb{R})$  satisfies

$$\lim_{y \rightarrow 0^+} \|f(\cdot + iy) - f\|_{L^p} = 0.$$

In particular,

$$\lim_{y \rightarrow 0^+} \int_{\mathbb{R}} |f(x + iy)|^p dx = \int_{\mathbb{R}} |f(x)|^p dx.$$

On the other hand, since  $F$  is analytic, the function  $|F|^p$  is subharmonic. Therefore the map

$$y \mapsto \int_{\mathbb{R}} |f(x + iy)|^p dx$$

is monotone decreasing as  $y \downarrow 0$ , and hence

$$\|f\|_{H^p}^p = \sup_{y>0} \int_{\mathbb{R}} |f(x + iy)|^p dx = \lim_{y \rightarrow 0^+} \int_{\mathbb{R}} |f(x + iy)|^p dx.$$

Combining the above identities, we conclude that

$$\|f\|_{L^p(\mathbb{R})}^p = \int_{\mathbb{R}} |f(x)|^p dx = \|f\|_{H^p}^p.$$

□

*Remark 2.20.* Then again, this result can be extended to the case  $H^{p>0}$  which is in fact a complete metric space, although it is not Banach for  $0 < p < 1$ . Since we are only concerned about the case  $p = 2$ , we prove this simplified version. In fact, the constant  $A_\alpha$  as the notation suggests can be made to depend only on  $\alpha$ , a detail that is irrelevant to our purposes.

### 2.2.2 Hilbert Transform is an Isometry in $L^2$ .

Our goal here is to introduce the Hilbert transform, show how it relates to Hardy spaces and prove an isometry result for  $L^2$ . In this moment we are going to use some properties of the Fourier transform, to avoid deviation we suggest a quick read on the chapter zero of [12].

**Definition 2.21** (Hilbert transform via the upper half-plane). Let  $f \in L^2(\mathbb{R})$  and let  $\mathcal{H}^2$  be the upper half-plane. Define the Poisson extension of  $f$  by

$$u(x, y) = (P_y * f)(x), \quad P_y(x) = \frac{1}{\pi} \frac{y}{x^2 + y^2}.$$

Then  $u$  is harmonic and satisfies

$$\sup_{y>0} \|u(\cdot, y)\|_{L^2} = \|f\|_{L^2}.$$

Let  $v(x, y)$  be the harmonic conjugate of  $u(x, y)$ , uniquely determined by the Cauchy–Riemann equations

$$\partial_x u = \partial_y v, \quad \partial_y u = -\partial_x v,$$

together with the normalization  $v(\cdot, y) \rightarrow 0$  as  $|x| \rightarrow \infty$ . Provided that it exists, the *Hilbert transform* of  $f$  is defined as the boundary value

$$Hf(x) = \lim_{y \rightarrow 0^+} v(x, y),$$

where the limit is taken in  $L^2(\mathbb{R})$ , and hence almost everywhere.

We shall show below that this limit exists for  $f \in \mathcal{S}(\mathbb{R})$  and extends uniquely to all of  $L^2(\mathbb{R})$ .

*Remark 2.22.* We would like to clarify why it actually holds “...Then  $u$  is harmonic and satisfies  $\sup_{y>0} \|u(\cdot, y)\|_{L^2} = \|f\|_{L^2}$ . ” Let  $f \in L^2(\mathbb{R})$  and define

$$u(x, y) = (P_y * f)(x), \quad P_y(x) = \frac{1}{\pi} \frac{y}{x^2 + y^2}.$$

Recall that the Fourier transform of the Poisson kernel is given by

$$\widehat{P_y}(\xi) = e^{-2\pi y|\xi|}.$$

Therefore,

$$\widehat{u(\cdot, y)}(\xi) = \widehat{P_y * f}(\xi) = e^{-2\pi y|\xi|} \hat{f}(\xi).$$

By Plancherel's theorem,

$$\begin{aligned} \|u(\cdot, y)\|_{L^2}^2 &= \int_{\mathbb{R}} \left| \widehat{u(\cdot, y)}(\xi) \right|^2 d\xi \\ &= \int_{\mathbb{R}} e^{-4\pi y|\xi|} |\hat{f}(\xi)|^2 d\xi. \end{aligned}$$

Since  $0 < e^{-4\pi y|\xi|} \leq 1$  for all  $\xi$ , it follows that

$$\|u(\cdot, y)\|_{L^2}^2 \leq \int_{\mathbb{R}} |\hat{f}(\xi)|^2 d\xi = \|f\|_{L^2}^2.$$

Hence,

$$\sup_{y>0} \|u(\cdot, y)\|_{L^2} \leq \|f\|_{L^2}.$$

On the other hand, as  $y \rightarrow 0^+$  we have

$$e^{-4\pi y|\xi|} \rightarrow 1 \quad \text{for all } \xi,$$

and the integrand is dominated by  $|\hat{f}(\xi)|^2 \in L^1(\mathbb{R})$ . By the dominated convergence theorem,

$$\lim_{y \rightarrow 0^+} \|u(\cdot, y)\|_{L^2}^2 = \int_{\mathbb{R}} |\hat{f}(\xi)|^2 d\xi = \|f\|_{L^2}^2.$$

Combining the two estimates yields

$$\sup_{y>0} \|u(\cdot, y)\|_{L^2} = \|f\|_{L^2},$$

which concludes.

**Lemma 2.23.** *Let  $f \in \mathcal{S}(\mathbb{R})$  and define*

$$v(x, y) := (Q_y * f)(x),$$

*where  $Q_y$  is the conjugate Poisson kernel. Then the limit*

$$\lim_{y \rightarrow 0^+} v(\cdot, y)$$

*exists in  $L^2(\mathbb{R})$ .*

*Proof.* Let  $f \in \mathcal{S}(\mathbb{R})$ . By Plancherel's theorem and the Fourier representation of the conjugate Poisson kernel, we have

$$\widehat{v(\cdot, y)}(\xi) = -i \operatorname{sgn}(\xi) e^{-2\pi y |\xi|} \hat{f}(\xi).$$

We first observe that

$$\|v(\cdot, y)\|_{L^2}^2 = \int_{\mathbb{R}} e^{-4\pi y |\xi|} |\hat{f}(\xi)|^2 d\xi \leq \|\hat{f}\|_{L^2}^2 = \|f\|_{L^2}^2,$$

so  $\{v(\cdot, y)\}_{y>0}$  is uniformly bounded in  $L^2(\mathbb{R})$ . Next, for  $y, y' > 0$ , we estimate

$$\|v(\cdot, y) - v(\cdot, y')\|_{L^2}^2 = \int_{\mathbb{R}} |e^{-2\pi y |\xi|} - e^{-2\pi y' |\xi|}|^2 |\hat{f}(\xi)|^2 d\xi.$$

For each fixed  $\xi \in \mathbb{R}$ ,

$$\lim_{y, y' \rightarrow 0^+} |e^{-2\pi y |\xi|} - e^{-2\pi y' |\xi|}|^2 = 0.$$

Moreover, since

$$|e^{-2\pi y |\xi|} - e^{-2\pi y' |\xi|}|^2 \leq 4,$$

we obtain the uniform bound

$$|e^{-2\pi y |\xi|} - e^{-2\pi y' |\xi|}|^2 |\hat{f}(\xi)|^2 \leq 4 |\hat{f}(\xi)|^2.$$

Because  $f \in \mathcal{S}(\mathbb{R})$ , we have  $\hat{f} \in \mathcal{S}(\mathbb{R}) \subset L^1(\mathbb{R})$ , hence  $|\hat{f}|^2 \in L^1(\mathbb{R})$ . Therefore, the dominated convergence theorem applies and yields

$$\lim_{y, y' \rightarrow 0^+} \|v(\cdot, y) - v(\cdot, y')\|_{L^2}^2 = 0.$$

Thus  $\{v(\cdot, y)\}_{y>0}$  is a Cauchy family in  $L^2(\mathbb{R})$ . Since  $L^2(\mathbb{R})$  is complete, there exists  $v_0 \in L^2(\mathbb{R})$  such that

$$v(\cdot, y) \rightarrow v_0 \quad \text{in } L^2(\mathbb{R}) \quad \text{as } y \rightarrow 0^+.$$

We set  $Hf = v_0$  and note that the Hilbert transform as we defined indeed exist for Schwartz functions. At this stage, the Hilbert transform is well-defined on  $\mathcal{S}(\mathbb{R})$ . Its extension to  $L^2(\mathbb{R})$  will follow from the isometry theorem.  $\square$

**Lemma 2.24** (Fourier transform of the Hilbert kernel). *Let  $\varphi \in \mathcal{S}(\mathbb{R})$  be a Schwartz function and define*

$$K(x) = \frac{1}{\pi} \text{p.v.} \frac{1}{x}.$$

Then

$$\int_{\mathbb{R}} (K * \varphi)(x) e^{-i\xi x} dx = -i \operatorname{sgn}(\xi) \hat{\varphi}(\xi),$$

that is,

$$\widehat{K * \varphi}(\xi) = -i \operatorname{sgn}(\xi) \hat{\varphi}(\xi) \quad \text{for all } \xi \in \mathbb{R}.$$

*Remark 2.25.* Here p.v. denotes the *Cauchy principal value*. Since the function  $x \mapsto 1/x$  is not locally integrable at the origin, the  $K$  is defined by the symmetric limit

$$\text{p.v.} \int_{\mathbb{R}} \frac{\varphi(x)}{x} dx := \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx, \quad \varphi \in \mathcal{S}(\mathbb{R}).$$

Equivalently, the kernel  $K$  acts on test functions by canceling the singularity at  $x = 0$  through symmetric integration.

*Proof.* We begin by defining the principal value convolution explicitly:

$$(K * \varphi)(x) = \frac{1}{\pi} \lim_{\varepsilon \rightarrow 0^+} \int_{|t| > \varepsilon} \frac{\varphi(x-t)}{t} dt.$$

Since  $\varphi$  is rapidly decreasing and smooth, the truncated integral is absolutely convergent and the limit exists uniformly in  $x$ . Taking the Fourier transform, we write

$$\int_{\mathbb{R}} (K * \varphi)(x) e^{-i\xi x} dx = \frac{1}{\pi} \lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}} \int_{|t| > \varepsilon} \frac{\varphi(x-t)}{t} e^{-i\xi x} dt dx.$$

By Fubini's theorem, the order of integration may be exchanged. Performing the change of variables  $y = x - t$ , we obtain

$$\int_{\mathbb{R}} (K * \varphi)(x) e^{-i\xi x} dx = \frac{1}{\pi} \lim_{\varepsilon \rightarrow 0^+} \int_{|t| > \varepsilon} \frac{e^{-i\xi t}}{t} \left( \int_{\mathbb{R}} \varphi(y) e^{-i\xi y} dy \right) dt.$$

Thus,

$$\int_{\mathbb{R}} (K * \varphi)(x) e^{-i\xi x} dx = \hat{\varphi}(\xi) \cdot \frac{1}{\pi} \lim_{\varepsilon \rightarrow 0^+} \int_{|t| > \varepsilon} \frac{e^{-i\xi t}}{t} dt.$$

We now compute the remaining integral. For  $\xi \neq 0$ ,

$$\int_{|t| > \varepsilon} \frac{e^{-i\xi t}}{t} dt = -2i \int_{\varepsilon}^{\infty} \frac{\sin(\xi t)}{t} dt.$$

Letting  $\varepsilon \rightarrow 0^+$  and using the classical identity

$$\int_0^\infty \frac{\sin(at)}{t} dt = \frac{\pi}{2} \operatorname{sgn}(a),$$

we obtain

$$\lim_{\varepsilon \rightarrow 0^+} \int_{|t| > \varepsilon} \frac{e^{-i\xi t}}{t} dt = -i\pi \operatorname{sgn}(\xi).$$

Substituting this back, we conclude that

$$\int_{\mathbb{R}} (K * \varphi)(x) e^{-i\xi x} dx = -i \operatorname{sgn}(\xi) \hat{\varphi}(\xi),$$

which proves the result.  $\square$

**Proposition 2.26** (Equivalence of the definitions of the Hilbert transform). *Let  $f \in L^2(\mathbb{R})$  and let*

$$u(x, y) = (P_y * f)(x), \quad v(x, y) = (Q_y * f)(x)$$

*be the Poisson extension of  $f$  to the upper half-plane and its harmonic conjugate. Then the boundary value of  $v$  satisfies*

$$\lim_{y \rightarrow 0} v(x, y) = Hf(x)$$

*in  $L^2(\mathbb{R})$ , where*

$$Hf(x) = \frac{1}{\pi} \text{p.v.} \int_{\mathbb{R}} \frac{f(x-t)}{t} dt$$

*is the Hilbert transform defined through the principal value kernel.*

*Proof.* By Lemma 2.23 we have

$$\widehat{v(\cdot, y)}(\xi) = -i \operatorname{sgn}(\xi) e^{-2\pi y |\xi|} \widehat{f}(\xi).$$

Letting  $y \rightarrow 0$  and using the dominated convergence theorem we obtain

$$\widehat{v}(\xi) = -i \operatorname{sgn}(\xi) \widehat{f}(\xi),$$

where  $v$  denotes the  $L^2$  limit of  $v(\cdot, y)$ . On the other hand, by Lemma 2.24 we have

$$\widehat{K * f}(\xi) = -i \operatorname{sgn}(\xi) \widehat{f}(\xi).$$

Since the Fourier transform is injective on  $L^2(\mathbb{R})$ , it follows that

$$v = K * f.$$

Consequently

$$v(x) = \frac{1}{\pi} \text{p.v.} \int_{\mathbb{R}} \frac{f(x-t)}{t} dt$$

for almost every  $x$ , which shows that the boundary value of the harmonic conjugate coincides with the principal value definition of the Hilbert transform.  $\square$

**Theorem 2.27** (Isometry of the Hilbert transform on  $L^2(\mathbb{R})$ ). *For every  $f \in L^2(\mathbb{R})$ ,*

$$\|Hf\|_2 = \|f\|_2.$$

*In particular, the Hilbert transform extends to an isometric operator on  $L^2(\mathbb{R})$ .*

*Proof.* The Hilbert transform can be written as a convolution

$$Hf = K * f, \quad K(x) = \frac{1}{\pi} \text{p.v.} \frac{1}{x}.$$

The Fourier transform of  $K$  is

$$\widehat{K}(\xi) = -i \operatorname{sgn}(\xi).$$

Hence,

$$\widehat{Hf}(\xi) = -i \operatorname{sgn}(\xi) \hat{f}(\xi).$$

By Plancherel's theorem,

$$\|Hf\|_2^2 = \int_{\mathbb{R}} |\widehat{Hf}(\xi)|^2 d\xi = \int_{\mathbb{R}} |-i \operatorname{sgn}(\xi) \hat{f}(\xi)|^2 d\xi.$$

Since  $|\operatorname{sgn}(\xi)| = 1$  almost everywhere, we obtain

$$\|Hf\|_2^2 = \int_{\mathbb{R}} |\hat{f}(\xi)|^2 d\xi = \|f\|_2^2.$$

The last equality being due to Plancherel again. This together with density of  $S(\mathbb{R})$  proves that the Hilbert transform is an isometry on  $L^2(\mathbb{R})$ .  $\square$

Next theorem puts everything together nicely .

**Theorem 2.28** (Hilbert transform and the  $H^2(dt)$  condition). *Let  $f \in L^2(\mathbb{R})$  and let*

$$u(x, y) = P_y * f(x), \quad v(x, y) = Q_y * f(x)$$

*be respectively the Poisson extension of  $f$  and its harmonic conjugate in the upper half-plane. Define*

$$F(x + iy) = u(x, y) + i v(x, y).$$

Then  $F$  belongs to the Hardy space  $H^2(dt)$ , that is,

$$\sup_{y>0} \int_{\mathbb{R}} |F(x + iy)|^2 dx < \infty.$$

Moreover, the boundary values satisfy

$$v(x) = Hf(x) \quad \text{a.e. on } \mathbb{R}.$$

*Proof.* Since  $F$  is holomorphic in  $\mathcal{H}^2$ , we have

$$|F(x + iy)|^2 = u(x, y)^2 + v(x, y)^2.$$

Integrating in  $x$  yields

$$\int_{\mathbb{R}} |F(x + iy)|^2 dx = \|u(\cdot, y)\|_2^2 + \|v(\cdot, y)\|_2^2.$$

By Plancherel's theorem,

$$\|u(\cdot, y)\|_2 \leq \|f\|_2 \quad \text{for all } y > 0.$$

Moreover, the harmonic conjugate kernel  $Q_y$  has Fourier multiplier  $-i \operatorname{sgn}(\xi)$ , hence

$$\|v(\cdot, y)\|_2 = \|Hf\|_2.$$

Therefore,

$$\sup_{y>0} \int_{\mathbb{R}} |F(x + iy)|^2 dx \leq \|f\|_2^2 + \|Hf\|_2^2 < \infty,$$

showing that  $F \in H^2(dt)$ .

Conversely, Theorem 2.11 implies that the nontangential boundary values of  $v$  exist almost everywhere and coincide with the Hilbert transform of  $f$ .  $\square$

## 2.3 The Two-Dimensional Case: A Complex-Analytic Proof via Outer Functions

In this section we finally prove Theorem 2.1, and use it to prove a uniqueness theorem to harmonic functions on the upper half plane, from which our main result will follow as a corollary.

### 2.3.1 The Key Characterization of Outer Functions

Theorem 2.1 tells us that the modulus of the boundary values of a non-zero  $H^p$  function *must* satisfy the logarithmic integrability condition (2.1). Violation of this condition forces the function to be identically zero. We split the proof of Theorem 2.1 in two parts. The following lemma will address the “necessity”.

**Lemma 2.29.** *If  $0 < p < \infty$  and if  $f(z) \in H^p(dt)$ ,  $f \not\equiv 0$  then*

$$\frac{1}{\pi} \int_{\mathbb{R}} \frac{\log |f(t)|}{1+t^2} dt > -\infty$$

*Proof.* Since  $f$  is holomorphic in  $\mathcal{H}^2$  and not identically zero, the function  $\log |f|$  is subharmonic in  $\mathcal{H}^2$ .

Since  $f \in H^p(dt)$ , the function  $\log |f|$  has integrable positive part uniformly in  $y$ . Indeed, for  $p > 0$ ,

$$\log^+ |f(z)| \leq \frac{1}{p} |f(z)|^p,$$

and therefore

$$\sup_{y>0} \int_{\mathbb{R}} \log^+ |f(x+iy)| dx \leq \frac{1}{p} \sup_{y>0} \int_{\mathbb{R}} |f(x+iy)|^p dx < \infty.$$

Let  $z = x + iy \in \mathcal{H}^2$ . Since  $\log |f(z)|$  is subharmonic and its positive part is uniformly integrable, and the Poisson Kernel is positive and integrable, by Lemma 2.18:

$$\begin{aligned} \log |f(x+iy)| &\leq \liminf_{r \rightarrow 0^+} \int_{\mathbb{R}} \log |f(t+ir)| P_y(x-t) dt \\ &= \frac{1}{\pi} \int_{\mathbb{R}} \log |f(t)| \frac{y}{(x-t)^2 + y^2} dt. \end{aligned}$$

On the previous equality, we pass  $\liminf$  under the integration sign after a Dominated Convergence Theorem argument. Since  $f$  is holomorphic and not identically zero, its zeros in  $\mathcal{H}^2$  are isolated. Hence there exists  $z_0$  such that  $f(z_0) \neq 0$ . Evaluating the inequality at such a point, and choosing  $z_0 = i$  for simplicity, we have

$$\log |f(i)| \leq \frac{1}{\pi} \int_{\mathbb{R}} \log |f(t)| \frac{1}{1+t^2} dt.$$

□

Now for the “sufficiency”:

**Lemma 2.30.** *If  $h(t)$  is a nonnegative function on  $L^p(\mathbb{R})$  such that (2.1) holds. Then there exists a  $f(z) \in H^p(dt)$  such that  $|f(t)| = h(t)$  almost everywhere.*

*Proof.* Since  $h \in L^p(\mathbb{R})$  and  $h \geq 0$ , we have the pointwise inequality

$$\log h(t) \leq \frac{1}{p} h(t)^p,$$

which implies that  $\log h$  is locally integrable and that (2.1) is equivalent to

$$\log h \in L^1\left(\frac{dt}{1+t^2}\right).$$

Let  $u(x, y)$  be the Poisson extension of  $\log h$  to the upper half-plane,

$$u(x, y) = \int_{\mathbb{R}} P_y(x-t) \log h(t) dt, \quad P_y(t) = \frac{1}{\pi} \frac{y}{t^2 + y^2}.$$

Then  $u$  is harmonic on  $\mathcal{H}^2$  and admits a harmonic conjugate  $v(x, y)$ , uniquely determined up to an additive constant. Define

$$f(z) = e^{u(z)+iv(z)}, \quad z = x + iy.$$

Since  $u + iv$  is holomorphic,  $f$  is holomorphic on  $\mathcal{H}^2$ . We now show that  $f \in H^p(dt)$ . For  $p < \infty$ , observe that

$$|f(x + iy)|^p = e^{pu(x,y)}.$$

By Jensen's inequality applied to the probability measure  $P_y(x-t) dt$ , we obtain

$$e^{pu(x,y)} = \exp\left(\int_{\mathbb{R}} P_y(x-t) p \log h(t) dt\right) \leq \int_{\mathbb{R}} P_y(x-t) h(t)^p dt.$$

Integrating in  $x$  and using  $\int_{\mathbb{R}} P_y(x-t) dx = 1$ , we get

$$\int_{\mathbb{R}} |f(x + iy)|^p dx \leq \int_{\mathbb{R}} h(t)^p dt.$$

Taking the supremum over  $y > 0$  yields

$$\sup_{y>0} \int_{\mathbb{R}} |f(x + iy)|^p dx \leq \|h\|_{L^p}^p < \infty,$$

so  $f \in H^p(dt)$ .

If  $p = \infty$ , then  $\log h \in L^\infty(\mathbb{R})$  implies that  $u(x, y)$  is bounded above, hence  $|f(z)| = e^{u(z)}$  is bounded and  $f \in H^\infty$ .

Finally, since  $u(x, y)$  converges nontangentially almost everywhere to  $\log h(x)$  as  $y \rightarrow 0^+$ , we obtain

$$|f(x)| = \lim_{y \rightarrow 0^+} e^{u(x,y)} = h(x) \quad \text{a.e. on } \mathbb{R}.$$

This completes the proof.  $\square$

### 2.3.2 Main Uniqueness Theorem for $d = 2$

We now state and prove the main result for the one-dimensional case.

**Theorem 2.31** (Simultaneous Vanishing on the Boundary). *Let  $F = u + iv \in H^p(\mathcal{H}^2)$  for some  $1 \leq p \leq \infty$ , and let  $E \subset \mathbb{R}$  be a measurable set with  $|E| > 0$ . Suppose that the nontangential boundary limits satisfy*

$$u(t) = 0 \quad \text{and} \quad v(t) = 0 \quad \text{for almost every } t \in E.$$

*Then  $F \equiv 0$  in  $\mathcal{H}^2$ .*

*Proof.* Assume, for contradiction, that  $F \not\equiv 0$ . Let  $h(t) = |F(t)|$  be the modulus of its boundary values. Since  $F \in H^p$ , we have  $h \in L^p(\mathbb{R})$ .

On the set  $E$ , the hypothesis  $u(t) = v(t) = 0$  implies  $F(t) = 0$  almost everywhere on  $E$ , and therefore  $h(t) = 0$  a.e. on  $E$ . Consequently,  $\log h(t) = -\infty$  on  $E$ .

Now consider the integral in the necessary condition (2.1):

$$\int_{\mathbb{R}} \frac{\log h(t)}{1+t^2} dt = \int_E \frac{\log h(t)}{1+t^2} dt + \int_{\mathbb{R} \setminus E} \frac{\log h(t)}{1+t^2} dt.$$

Since  $|E| > 0$ , the integral over  $E$  equals  $-\infty$ . Even if the integral over  $\mathbb{R} \setminus E$  is finite, the total integral is  $-\infty$ :

$$\int_{\mathbb{R}} \frac{\log h(t)}{1+t^2} dt = -\infty.$$

However, by Theorem 2.1, the boundary modulus  $h(t)$  of any non-zero  $H^p$  function must satisfy (2.1). The equality above directly contradicts that condition. Therefore, our initial assumption  $F \not\equiv 0$  is false, and we conclude that  $F \equiv 0$ .  $\square$

The elegance of this proof lies in its direct use of the fundamental characterization of outer functions. The moment a non-zero  $H^p$  function vanishes on a set of positive measure, its boundary modulus forces the logarithmic integral to be  $-\infty$ , an impossibility.

### 2.3.3 Application to the Harmonic Problem

We now translate Theorem 2.31 into the language of harmonic functions and their normal derivatives, which is the original formulation of our problem.

**Corollary 2.32** (Vanishing of Function and Normal Derivative). *Let  $u$  be a real-valued harmonic function in  $\mathcal{H}^2 = \mathbb{R} \times (0, \infty)$  whose boundary trace  $f(x) = u(x, 0)$  (defined nontangentially a.e.) belongs to  $L^2(\mathbb{R})$ . Let  $\partial_y u$  denote its normal derivative. If there exists a set  $A \subset \mathbb{R}$  with  $|A| > 0$  such that*

$$u(x, 0) = 0 \quad \text{and} \quad \partial_y u(x, 0) = 0 \quad \text{for almost every } x \in A,$$

*then  $u \equiv 0$  in  $\mathcal{H}^2$ . In particular,*

$$|\{x \in \mathbb{R} : u(x, 0) = 0 \text{ and } \partial_y u(x, 0) = 0\}| = 0.$$

*Proof.* Since  $f \in L^2(\mathbb{R})$ , its harmonic extension  $u$  is the Poisson integral of  $f$ . Let  $v$  be the harmonic conjugate of  $u$  in  $\mathcal{H}^2$ , normalized so that  $v(x, 0) \in L^2(\mathbb{R})$  (this is possible because the Hilbert transform maps  $L^2$  to  $L^2$ ). Then, by Theorem 2.28  $F = u + iv$  belongs to  $H^2(dt)$ .

The Cauchy–Riemann equations imply that  $\partial_y u = -\partial_x v$ . Since  $u$  and  $v$  are Poisson integrals of their boundary values, the condition  $\partial_y u(x, 0) = 0$  on  $A$  implies that the boundary function  $v(x, 0)$  is locally constant on  $A$ . Because  $u(x, 0) = 0$  on  $A$ , we may adjust  $v$  by an additive constant (which does not affect  $F$  being in  $H^2$ ) to obtain  $v(x, 0) = 0$  on  $A$ . Thus, both the real and imaginary parts of  $F$  vanish on  $A$ . By Theorem 2.31,  $F \equiv 0$ , hence  $u \equiv 0$ .  $\square$

*Remark 2.33.* The condition  $f \in L^2(\mathbb{R})$  is essential. There exist non-zero analytic functions in  $\mathcal{H}^2$  whose real and imaginary parts vanish on an interval (e.g.,  $e^{-1/z^2}$ -type examples on a line), but they do not belong to any Hardy space because their boundary values are not in  $L^p$ . The  $H^p$  framework provides the precise control needed to enforce uniqueness.

# The case $\mathcal{H}^d, d \geq 3$

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In this chapter, we prove the following result:

**Theorem 3.1.** *If  $d \geq 3$  there exists a harmonic function  $f : \mathcal{H}^d \rightarrow \mathbb{R}$ ,  $C^1$  up to the boundary, and such that  $f$  and  $\nabla f$  vanish on a common set of the boundary, with positive measure.*

## 3.1 Some Harmonic Analysis

### 3.1.1 Poisson Kernels Again

In the previous chapter we used extensively that if  $f \in L^p(\mathbb{R})$ , then  $u(x, y) := (P_y * f)(x)$  is harmonic on the upper half plane. We had made a heuristic discussion of this fact, but no proof was present.

Taking this as an opportunity to introduce the Poisson Kernel on the upper half space, we finally prove this fact.

**Definition 3.2.** Let  $d \geq 2$  and let  $\mathcal{H}^d$  be the upper half space. The Poisson kernel in  $\mathcal{H}^d$  is defined by

$$P_y(x) = c_d \frac{y}{(|x|^2 + y^2)^{\frac{d}{2}}}, \quad x \in \mathbb{R}^{d-1}, y > 0,$$

where the constant  $c_d$  is chosen so that

$$\int_{\mathbb{R}^{d-1}} P_y(x) dx = 1.$$

**Lemma 3.3.** *The Poisson kernel*

$$P_y(x) = c_d \frac{y}{(|x|^2 + y^2)^{\frac{d}{2}}}, \quad (x, y) \in \mathcal{H}^d$$

*is harmonic in  $\mathcal{H}^d$ , i.e.*

$$\Delta P_y(x) = 0.$$

*Proof.* Let  $r^2 = |x|^2 + y^2$ . Then

$$P_y(x) = c_d y r^{-d}.$$

We compute the Laplacian of  $y r^{-d}$ . Since

$$\partial_{x_j}(r^{-d}) = -d x_j r^{-d-2},$$

we obtain

$$\partial_{x_j}(y r^{-d}) = -d y x_j r^{-d-2}.$$

Differentiating again,

$$\partial_{x_j}^2(y r^{-d}) = -d y r^{-d-2} + d(d+2) y x_j^2 r^{-d-4}.$$

Summing over  $j = 1, \dots, d-1$  gives

$$\sum_{j=1}^{d-1} \partial_{x_j}^2(y r^{-d}) = -d(d-1) y r^{-d-2} + d(d+2) y |x|^2 r^{-d-4}.$$

Next,

$$\partial_y(y r^{-d}) = r^{-d} - d y^2 r^{-d-2}$$

and

$$\partial_y^2(y r^{-d}) = -3d y r^{-d-2} + d(d+2) y^3 r^{-d-4}.$$

Adding the derivatives yields

$$\Delta(y r^{-d}) = -d(d+2) y r^{-d-2} + d(d+2) y r^2 r^{-d-4} = 0.$$

Therefore  $\Delta P_y(x) = 0$ . □

**Proposition 3.4.** *Let  $1 \leq p \leq \infty$  and let  $f \in L^p(\mathbb{R}^{d-1})$ . Define*

$$u(x, y) = (P_y * f)(x) = \int_{\mathbb{R}^{d-1}} P_y(x-t) f(t) dt, \quad (x, y) \in \mathcal{H}^d.$$

*Then,  $u$  is harmonic in  $\mathcal{H}^d$ , i.e.*

$$\Delta u = 0.$$

*Proof.* For each  $y > 0$  the function  $P_y$  belongs to  $L^1(\mathbb{R}^{d-1})$  and satisfies

$$\int_{\mathbb{R}^{d-1}} P_y(x) dx = 1.$$

Hence for  $f \in L^p(\mathbb{R}^{d-1})$  the convolution

$$u(x, y) = \int_{\mathbb{R}^{d-1}} P_y(x - t) f(t) dt$$

is well defined for every  $(x, y) \in \mathcal{H}^d$ .

Since  $P_y$  is  $C^2$  and belongs to  $L^1(\mathbb{R}^{d-1})$ , we may differentiate under the integral sign to obtain

$$\Delta u(x, y) = \int_{\mathbb{R}^{d-1}} (\Delta P_y)(x - t) f(t) dt = 0.$$

Hence  $u$  is harmonic in  $\mathcal{H}^d$ . □

*Remark 3.5.* A analogous for Theorem 2.4 still holds, but it won't play any role in this chapter.

### 3.1.2 The Riesz Transforms

The Hilbert transform plays a fundamental role in one-dimensional harmonic analysis. In higher dimensions, its natural analogues are the *Riesz transforms*. These operators arise naturally in the study of harmonic functions, singular integrals, and partial differential equations.

**Definition 3.6.** Let  $d \geq 2$ . For  $f \in \mathcal{S}(\mathbb{R}^{d-1})$  and  $1 \leq j \leq d - 1$ , the  $j$ -th Riesz transform of  $f$  is defined by

$$R_j f(x) = c_{d-1} \text{p.v.} \int_{\mathbb{R}^{d-1}} \frac{x_j - y_j}{|x - y|^d} f(y) dy,$$

where  $c_{d-1}$  is a dimensional constant and the integral is understood in the principal value sense.

Equivalently, the Riesz transforms can be defined through the Fourier transform.

**Proposition 3.7** ([13] Proposition 4.1.14). *For  $f \in \mathcal{S}(\mathbb{R}^d)$ ,*

$$\widehat{R_j f}(\xi) = -i \frac{\xi_j}{|\xi|} \hat{f}(\xi).$$

*Thus each Riesz transform is a Fourier multiplier operator whose symbol is*

$$m_j(\xi) = -i \frac{\xi_j}{|\xi|}.$$

**Corollary 3.8.** *The Riesz transform  $R_j$  is bounded on  $L^2(\mathbb{R}^d)$ .*

*Proof.* From the Fourier multiplier representation

$$\widehat{R_j f}(\xi) = -i \frac{\xi_j}{|\xi|} \hat{f}(\xi),$$

Plancherel's theorem gives

$$\|R_j f\|_2^2 = \int_{\mathbb{R}^d} \left| \frac{\xi_j}{|\xi|} \right|^2 |\hat{f}(\xi)|^2 d\xi.$$

Since

$$\left| \frac{\xi_j}{|\xi|} \right| \leq 1,$$

out of zero, we obtain

$$\|R_j f\|_2^2 \leq \int |\hat{f}(\xi)|^2 d\xi = \|f\|_2^2.$$

Therefore

$$\|R_j f\|_2 \leq \|f\|_2.$$

□

The Riesz transforms generalize the Hilbert transform to higher dimensions. Indeed, when  $d = 1$  there is only one transform and the formula above becomes

$$\widehat{Rf}(\xi) = -i \operatorname{sgn}(\xi) \hat{f}(\xi),$$

which we refer to Lemma 2.24 to see that it is precisely the Fourier multiplier of the Hilbert transform.

Hence the Hilbert transform may be viewed as the one-dimensional Riesz transform. Geometrically, the vector-valued operator

$$Rf = (R_1 f, \dots, R_d f)$$

can be interpreted as a directional derivative of the Newtonian potential. More precisely, if

$$\Phi(x) = c_d |x|^{2-d}$$

is the fundamental solution of the Laplacian in  $\mathbb{R}^d$ , then

$$R_j f = \partial_{x_j} (\Phi * f).$$

From the point of view of harmonic extensions, the Riesz transforms arise naturally from the Poisson kernel in the upper half space. If  $f$  is a function on  $\mathbb{R}^{d-1}$  and

$$u(x, y) = (P_y * f)(x)$$

is its Poisson extension to  $\mathcal{H}^d$ , then the tangential derivatives of  $u$  satisfy

$$\partial_{x_j} u(x, y) = (P_y * R_j f)(x).$$

Thus the Riesz transforms may be interpreted as boundary operators associated with harmonic functions in the upper half space, in the same way that the Hilbert transform arises as the boundary value of the harmonic conjugate in dimension one. Regarding the previous considerations we prove a closely related result, which will have an important role later on.

**Theorem 3.9.** *Let  $f \in \mathcal{S}(\mathbb{R}^{d-1})$  and define the harmonic extension*

$$u(x', x_d) = P_{x_d} * f(x')$$

where

$$P_t(x') = c_d \frac{t}{(|x'|^2 + t^2)^{d/2}}.$$

Then

$$\partial_{x_d} u(x', 0) = \sum_{k=1}^{d-1} R_k(\partial_{x_k} f)(x').$$

*Proof.* We use the Fourier transform

$$\widehat{f}(\xi) = \int_{\mathbb{R}^{d-1}} f(x') e^{-2\pi i x' \cdot \xi} dx'.$$

Then

$$\widehat{\partial_{x_k} f}(\xi) = 2\pi i \xi_k \widehat{f}(\xi).$$

The Fourier transform of the Poisson kernel satisfies

$$\widehat{P}_t(\xi) = e^{-2\pi t |\xi|}.$$

Hence

$$\widehat{u}(\xi, x_d) = e^{-2\pi x_d |\xi|} \widehat{f}(\xi).$$

Differentiating gives

$$\partial_{x_d} \widehat{u}(\xi, x_d) = -2\pi |\xi| e^{-2\pi x_d |\xi|} \widehat{f}(\xi).$$

Taking  $x_d \rightarrow 0$  yields

$$\widehat{\partial_{x_d} u}(\xi, 0) = -2\pi |\xi| \widehat{f}(\xi).$$

We recall the relation

$$\widehat{R_k g}(\xi) = -i \frac{\xi_k}{|\xi|} \widehat{g}(\xi).$$

from Proposition 3.7. Thus

$$\widehat{R_k(\partial_k f)}(\xi) = -i \frac{\xi_k}{|\xi|} (2\pi i \xi_k) \widehat{f}(\xi) = 2\pi \frac{\xi_k^2}{|\xi|} \widehat{f}(\xi).$$

Summing over  $k$  gives

$$\sum_{k=1}^{d-1} \widehat{R_k(\partial_k f)}(\xi) = 2\pi \frac{\sum \xi_k^2}{|\xi|} \widehat{f}(\xi) = 2\pi |\xi| \widehat{f}(\xi).$$

Comparing the two expressions shows

$$\partial_{x_d} u(x', 0) = - \sum_{k=1}^{d-1} R_k(\partial_k f)(x').$$

Since  $\partial_n u(x', 0) = -\partial_{x_d} u(x', 0)$ , we have  $\partial_n u(x', 0) = \sum_{k=1}^{d-1} R_k(\partial_k f)(x')$ . Denoting  $R(\nabla f) = \sum_{k=1}^{d-1} R_k(\partial_k f)$  we then obtain the short formula

$$\partial_n u = R(\nabla f)$$

□

**Corollary 3.10.** *Let  $f \in \mathcal{S}(\mathbb{R}^{d-1})$  and let*

$$u(x', x_d) = P_{x_d} * f(x')$$

*be its harmonic extension to the upper half-space. Then for each  $j = 1, \dots, d-1$  one has*

$$\partial_{x_j} u(x', 0) = -R_j(\partial_n u)(x').$$

*Consequently,*

$$|\nabla_{x'} u(x', 0)| \leq C |\partial_n u(x', 0)|.$$

*Proof.* From the proof of Theorem 3.9 we obtained

$$\widehat{\partial_{x_d} u}(\xi, 0) = -2\pi |\xi| \widehat{f}(\xi).$$

Since the outward normal derivative is

$$\partial_n u(x', 0) = -\partial_{x_d} u(x', 0),$$

we have

$$\widehat{\partial_n u}(\xi) = 2\pi |\xi| \widehat{f}(\xi).$$

Now consider the tangential derivative. Since

$$u(x', 0) = f(x'),$$

we obtain

$$\partial_{x_j} u(x', 0) = \partial_{x_j} f(x').$$

Taking Fourier transforms,

$$\widehat{\partial_{x_j} u}(\xi, 0) = 2\pi i \xi_j \widehat{f}(\xi).$$

Using the previous identity for  $\widehat{\partial_n u}$  we write

$$\widehat{\partial_{x_j} u}(\xi, 0) = i \frac{\xi_j}{|\xi|} \widehat{\partial_n u}(\xi).$$

Recalling the Fourier multiplier representation of the Riesz transform,

$$\widehat{R_j g}(\xi) = -i \frac{\xi_j}{|\xi|} \widehat{g}(\xi),$$

we obtain

$$\widehat{\partial_{x_j} u}(\xi, 0) = -\widehat{R_j(\partial_n u)}(\xi).$$

Taking the inverse Fourier transform yields

$$\partial_{x_j} u(x', 0) = -R_j(\partial_n u)(x').$$

Finally,

$$\nabla_{x'} u = (-R_1(\partial_n u), \dots, -R_{d-1}(\partial_n u)).$$

Hence

$$|\nabla_{x'} u(x', 0)| \leq C \sum_{j=1}^{d-1} |R_j(\partial_n u)(x')| \leq C |\partial_n u(x', 0)|$$

up to dimensional constants, which proves the claim.  $\square$

**Theorem 3.11.** *Let  $0 < \alpha < 1$  and let  $f \in C^\alpha(\mathbb{R}^d)$ . Let  $R_k$  be the  $k$ -th Riesz transform*

$$R_k f(x) = \text{p.v.} \int_{\mathbb{R}^d} \frac{x_k - y_k}{|x - y|^{d+1}} f(y) dy.$$

*Then*

$$|R_k f(x) - R_k f(z)| \leq C \|f\|_{C^\alpha} |x - z|^\alpha.$$

*Proof.* Let

$$K(z) = \frac{z_k}{|z|^{d+1}}.$$

Then

$$R_k f(x) = \text{p.v.} \int K(x - y) f(y) dy.$$

We claim

$$\int_{|z|=r} K(z) d\sigma(z) = 0.$$

On  $|z| = r$ ,

$$K(z) = \frac{z_k}{r^{d+1}}.$$

Thus

$$\int_{|z|=r} K(z) d\sigma(z) = \frac{1}{r^{d+1}} \int_{|z|=r} z_k d\sigma(z).$$

The reflection

$$(z_1, \dots, z_k, \dots, z_d) \mapsto (z_1, \dots, -z_k, \dots, z_d)$$

preserves the sphere and the surface measure. Hence

$$\int_{|z|=r} z_k d\sigma(z) = - \int_{|z|=r} z_k d\sigma(z),$$

so the integral is zero.

Using the principal value definition,

$$R_k f(x) = \lim_{\varepsilon \rightarrow 0} \int_{|x-y|>\varepsilon} K(x-y) f(y) dy.$$

Write

$$f(y) = f(y) - f(x) + f(x).$$

Thus

$$\begin{aligned} \int_{|x-y|>\varepsilon} K(x-y) f(y) dy &= \int_{|x-y|>\varepsilon} K(x-y) (f(y) - f(x)) dy + \\ &\quad + f(x) \int_{|x-y|>\varepsilon} K(x-y) dy. \end{aligned}$$

By the cancellation property, the second term vanishes. Hence

$$R_k f(x) = \int K(x-y) (f(y) - f(x)) dy.$$

Let  $x, z \in \mathbb{R}^d$ . Then

$$\begin{aligned} R_k f(x) - R_k f(z) &= \int (K(x-y) - K(z-y)) (f(y) - f(x)) dy + \\ &\quad + \int K(z-y) (f(x) - f(z)) dy. \end{aligned}$$

Since  $f \in C^\alpha$ ,

$$|f(y) - f(x)| \leq \|f\|_{C^\alpha} |x - y|^\alpha.$$

Split the integral into the regions  $|x - y| \leq 2|x - z|$  and  $|x - y| > 2|x - z|$ .

**Inside region.**

Let  $\delta = |x - z|$ . Consider

$$I_1 = \int_{|x-y| \leq 2\delta} |K(x-y)| |f(y) - f(x)| dy.$$

Since

$$|K(x-y)| \leq C|x-y|^{-d}$$

and

$$|f(y) - f(x)| \leq [f]_{C^\alpha} |x - y|^\alpha,$$

we obtain

$$I_1 \leq C[f]_{C^\alpha} \int_{|x-y| \leq 2\delta} |x-y|^{\alpha-d} dy.$$

Changing variables  $u = x - y$ ,

$$I_1 \leq C[f]_{C^\alpha} \int_{|u| \leq 2\delta} |u|^{\alpha-d} du.$$

Using polar coordinates,

$$I_1 \leq C[f]_{C^\alpha} \int_0^{2\delta} r^{\alpha-d} r^{d-1} dr = C[f]_{C^\alpha} \int_0^{2\delta} r^{\alpha-1} dr.$$

Thus

$$I_1 \leq C[f]_{C^\alpha} \delta^\alpha.$$

**Outside region.**

Consider

$$I_2 = \int_{|x-y| > 2\delta} |K(x-y) - K(z-y)| |f(y) - f(x)| dy.$$

The kernel satisfies

$$|\nabla K(w)| \leq C|w|^{-d-1}.$$

Hence by the mean value theorem

$$|K(x-y) - K(z-y)| \leq C \frac{|x-z|}{|x-y|^{d+1}}.$$

Thus

$$I_2 \leq C[f]_{C^\alpha} \delta \int_{|x-y|>2\delta} |x-y|^{\alpha-d-1} dy.$$

Let  $u = x - y$ . Then

$$I_2 \leq C[f]_{C^\alpha} \delta \int_{|u|>2\delta} |u|^{\alpha-d-1} du.$$

Using polar coordinates,

$$I_2 \leq C[f]_{C^\alpha} \delta \int_{2\delta}^{\infty} r^{\alpha-d-1} r^{d-1} dr = C[f]_{C^\alpha} \delta \int_{2\delta}^{\infty} r^{\alpha-2} dr.$$

Since  $0 < \alpha < 1$ ,

$$\int_{2\delta}^{\infty} r^{\alpha-2} dr = C\delta^{\alpha-1}.$$

Therefore

$$I_2 \leq C[f]_{C^\alpha} \delta^\alpha.$$

Combining the estimates yields

$$|R_k f(x) - R_k f(z)| \leq C\|f\|_{C^\alpha} |x - z|^\alpha.$$

□

### 3.1.3 Calderón-Zygmund Decomposition

Still concerning Riesz transforms we are going to need an important embedding result; that  $R$  maps  $L^1$  to  $L^{1,\infty}$ . In order to do so, the Calderón-Zygmund decomposition theorem will play a major role.

The Calderón-Zygmund decomposition is a fundamental tool in harmonic analysis, particularly in the theory of singular integrals. It provides a systematic way to decompose an arbitrary integrable function into two carefully constructed components: a "good" bounded part and a "bad" oscillatory part.

Recall that a dyadic cube in  $\mathbb{R}^n$  is a set of the form

$$Q = [2^k m_1, 2^k(m_1 + 1)) \times [2^k m_2, 2^k(m_2 + 1)) \times \cdots \times [2^k m_n, 2^k(m_n + 1))$$

where  $k, m_1, m_2, \dots, m_n \in \mathbb{Z}$ . The integer  $k$  determines the scale (size) of the cube, and the integers  $m_1, \dots, m_n$  determine its position.

Some key properties of dyadic cubes: two dyadic cubes are either disjoint or one is contained in the other; the collection of all dyadic cubes at a fixed scale  $k$  partitions  $\mathbb{R}^n$ ; each dyadic cube at scale  $k$  can be subdivided into  $2^n$  dyadic cubes at scale  $k - 1$ .

For more understanding of dyadic cubes we recommend a reading on [14].

**Theorem 3.12** ([13] Theorem 4.3.1). *Let  $f \in L^1(\mathbb{R}^n)$  and  $\alpha > 0$ . Then there exist functions  $g$  and  $b$  on  $\mathbb{R}^n$  such that:*

1.  $f = g + b$
2.  $\|g\|_{L^1} \leq \|f\|_{L^1}$  and  $\|g\|_{L^\infty} \leq 2^n \alpha$
3.  $b = \sum_j b_j$ , where each  $b_j$  is supported in a dyadic cube  $Q_j$ . Furthermore, the cubes  $Q_j$  are pairwise disjoint
4.  $\int_{Q_j} b_j(x) dx = 0$  for each  $j$
5.  $\|b_j\|_{L^1} \leq 2^{n+1} \alpha |Q_j|$
6.  $\sum_j |Q_j| \leq \alpha^{-1} \|f\|_{L^1}$

The function  $g$  is called the good function as it is both integrable and bounded, while  $b$  is called the bad function as it contains the singular part of  $f$  but is carefully constructed to have mean value zero on each cube. Moreover, from properties (1) and (2), we obtain the estimate  $\|b\|_{L^1} \leq 2\|f\|_{L^1}$ .

The Calderón–Zygmund decomposition is called a stopping-time argument because it involves an iterative process where we examine dyadic cubes at different scales and “stop” the decomposition when a certain condition is met. The algorithm proceeds as follows:

1. Start with a dyadic cube and compute the average value of  $|f|$  over that cube;
2. If  $\frac{1}{|Q|} \int_Q |f(x)| dx \leq \alpha$ , continue to its dyadic children (subdivide);
3. If  $\frac{1}{|Q|} \int_Q |f(x)| dx > \alpha$ , stop the subdivision and designate  $Q$  as one of the cubes where the singular part will be supported;
4. This creates a collection of maximal dyadic cubes where the stopping condition is triggered.

**Theorem 3.13.** *The Riesz transform satisfies*

$$R_j : L^1(\mathbb{R}^d) \rightarrow L^{1,\infty}(\mathbb{R}^d).$$

*More precisely,*

$$|\{x : |R_j f(x)| > \lambda\}| \leq \frac{C}{\lambda} \|f\|_{L^1}.$$

*Proof.* Let  $f \in L^1(\mathbb{R}^d)$  and  $\lambda > 0$ . Using the Calderón–Zygmund decomposition at height  $\lambda$ , we write

$$f = g + b,$$

where

$$b = \sum_k b_k.$$

From the decomposition we have

$$\|g\|_\infty \leq C\lambda, \quad \|g\|_1 \leq \|f\|_1,$$

and each  $b_k$  is supported on a cube  $Q_k$  satisfying

$$\int b_k = 0, \quad \sum_k |Q_k| \leq \frac{1}{\lambda} \|f\|_1.$$

Since  $\|g\|_\infty \leq C\lambda$ , we obtain

$$\|g\|_2^2 = \int |g|^2 \leq \|g\|_\infty \int |g| \leq C\lambda \|f\|_1.$$

Since  $R_j$  is bounded on  $L^2$  (Lemma 3.8),

$$\|R_j g\|_2 \leq C \|g\|_2.$$

By Chebyshev's inequality,

$$|\{|R_j g| > \lambda/2\}| \leq \frac{C}{\lambda^2} \|R_j g\|_2^2 \leq \frac{C}{\lambda^2} \|g\|_2^2 \leq \frac{C}{\lambda} \|f\|_1.$$

Let

$$\Omega = \bigcup_k 2Q_k.$$

Since the cubes  $Q_k$  are disjoint,

$$|\Omega| \leq C \sum_k |Q_k| \leq \frac{C}{\lambda} \|f\|_1.$$

For  $x \notin 2Q_k$ , using the cancellation  $\int b_k = 0$  we write

$$R_j b_k(x) = \int (K_j(x-y) - K_j(x-c_k)) b_k(y) dy,$$

where  $c_k$  is the center of  $Q_k$ . The kernel

$$K_j(x) = c_d \frac{x_j}{|x|^{d+1}}$$

is smooth on  $\mathbb{R}^d \setminus \{0\}$  and readily satisfies

$$|\nabla K_j(x)| \leq \frac{C}{|x|^{d+1}}.$$

Indeed, by the mean value theorem,

$$|K_j(x-y) - K_j(x-c_k)| \leq |y-c_k| \sup_z |\nabla K_j(z)|,$$

where  $z$  lies on the segment joining  $x-y$  and  $x-c_k$ . If  $x \notin 2Q_k$  and  $y \in Q_k$ , the following lemma, Lemma 3.14, implies  $|z| \sim |x-c_k|$ , hence

$$|K_j(x-y) - K_j(x-c_k)| \leq C \frac{|y-c_k|}{|x-c_k|^{d+1}}.$$

Therefore

$$|R_j b_k(x)| \leq C \frac{\ell(Q_k)}{|x-c_k|^{d+1}} \|b_k\|_1,$$

where  $\ell(Q_k)$  denotes the side length of  $Q_k$ . Integrating over  $x \notin 2Q_k$  and summing in  $k$  yields

$$\int_{\Omega^c} |R_j b(x)| dx \leq C \|f\|_1.$$

Indeed,

$$\int_{(2Q_k)^c} |R_j b_k(x)| dx \leq C \|b_k\|_1 \ell(Q_k) \int_{|x-c_k| \gtrsim \ell(Q_k)} \frac{dx}{|x-c_k|^{d+1}}.$$

Using the change of variables  $x-c_k = \ell(Q_k)u$ , we obtain

$$\int_{|x-c_k| \gtrsim \ell(Q_k)} \frac{dx}{|x-c_k|^{d+1}} = \ell(Q_k)^d \int_{|u| \gtrsim 1} \frac{du}{(\ell(Q_k)|u|)^{d+1}} = \frac{1}{\ell(Q_k)} \int_{|u| \gtrsim 1} \frac{du}{|u|^{d+1}}.$$

Since the last integral is finite, it follows that

$$\int_{(2Q_k)^c} |R_j b_k(x)| dx \leq C \|b_k\|_1.$$

Summing over  $k$  yields

$$\int_{\Omega^c} |R_j b(x)| dx \leq \sum_k \int_{(2Q_k)^c} |R_j b_k(x)| dx \leq C \sum_k \|b_k\|_1.$$

From the Calderón–Zygmund decomposition we know that

$$\sum_k \|b_k\|_1 \leq C \|f\|_1,$$

and therefore

$$\int_{\Omega^c} |R_j b(x)| dx \leq C \|f\|_1.$$

By Chebyshev's inequality,

$$|\{x \notin \Omega : |R_j b(x)| > \lambda/2\}| \leq \frac{C}{\lambda} \|f\|_1.$$

Let

$$E = \{x : |R_j f(x)| > \lambda\}.$$

Since  $f = g + b$ , we have

$$R_j f = R_j g + R_j b,$$

and therefore

$$E \subset \{|R_j g| > \lambda/2\} \cup \{|R_j b| > \lambda/2\}.$$

Hence

$$|E| \leq |\{|R_j g| > \lambda/2\}| + |\{|R_j b| > \lambda/2\}|.$$

Moreover,

$$|\{|R_j b| > \lambda/2\}| \leq |\Omega| + |\{|R_j b| > \lambda/2\} \cap \Omega^c|.$$

Combining the estimates obtained above gives

$$|\{|R_j f| > \lambda\}| \leq \frac{C}{\lambda} \|f\|_1.$$

□

**Lemma 3.14.** *Let  $Q_k$  be a cube with center  $c_k$  and side length  $\ell(Q_k)$ . If  $x \notin 2Q_k$  and  $y \in Q_k$ , then*

$$|x - y| \sim |x - c_k|.$$

*Proof.* Since  $y \in Q_k$ , we have

$$|y - c_k| \leq \frac{\sqrt{d}}{2} \ell(Q_k).$$

If  $x \notin 2Q_k$ , then  $|x - c_k| > \ell(Q_k)$ . Using the triangle inequality we obtain

$$|x - y| \geq |x - c_k| - |y - c_k|$$

and

$$|x - y| \leq |x - c_k| + |y - c_k|.$$

Substituting the bound for  $|y - c_k|$  yields constants  $C_1, C_2 > 0$  such that

$$C_1|x - c_k| \leq |x - y| \leq C_2|x - c_k|.$$

□

## 3.2 The Three-Dimensional Case

Now we start with the construction of our function. Some technical lemmas are needed.

### 3.2.1 The Correction Function

**Lemma 3.15.** *Consider the functions  $G_\varepsilon : \mathcal{H}^d \rightarrow \mathbb{R}$  given by  $G_\varepsilon(x) = \frac{-(\varepsilon + x_d)}{|x + \varepsilon e_d|^d}$ . Then, for small enough  $p$  and  $\varepsilon$ , it holds that*

$$\int_{\mathbb{R}^{d-1}} \left( \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p - 1 \right) < -2M < 0, \quad (3.1)$$

where  $M > 0$  is independent of  $\varepsilon$ .

This lemma is due to Aleksandrov and Kargaev, its proof is beyond our interest and it is contained at a more general setting on [15].

It is assumed that anyone who has read this text so far is familiar with the construction of bump functions. However, we will need to introduce dyadic partitioning, and we take this opportunity to remind the reader. In this sense, the next lemma can be ignored if the technical details of this construction are not of your interest.

**Lemma 3.16.** *For  $j = 0, 1, \dots$  there exists  $\psi_j : \mathbb{R}^{d-1} \rightarrow \mathbb{R}$  smooth functions supported on the annuli  $\{2^{j-1}\delta \leq |x| \leq 2^{j+1}\delta\}$  such that,  $\sum_j \psi_j = 1$  for  $|x| > \delta$  and  $|\nabla^k \psi_j| \lesssim (2^j \delta)^{-k}$ .*

*Proof.* Define the smooth function

$$\rho(t) = \begin{cases} e^{-1/t} & t > 0, \\ 0 & t \leq 0. \end{cases}$$

Set

$$\theta(t) = \frac{\rho(t)}{\rho(t) + \rho(1-t)}.$$

Then  $\theta \in C^\infty(\mathbb{R})$ ,  $0 \leq \theta \leq 1$ , and

$$\theta(t) = 0 \text{ for } t \leq 0, \quad \theta(t) = 1 \text{ for } t \geq 1.$$

Using this function define the radial cutoff

$$\eta(x) = 1 - \theta(|x| - 1), \quad x \in \mathbb{R}^d.$$

Then  $\eta \in C_c^\infty(\mathbb{R}^d)$  and

$$\eta(x) = \begin{cases} 1 & |x| \leq 1, \\ 0 & |x| \geq 2, \end{cases} \quad 0 \leq \eta \leq 1.$$

Let  $\delta > 0$  be fixed and define

$$\psi_0(x) = \eta(x/\delta),$$

and for  $j \geq 1$

$$\psi_j(x) = \eta\left(\frac{x}{2^j\delta}\right) - \eta\left(\frac{x}{2^{j-1}\delta}\right).$$

For  $j \geq 1$  we have

$$\psi_j(x) \neq 0$$

only when

$$\eta\left(\frac{x}{2^j\delta}\right) \neq \eta\left(\frac{x}{2^{j-1}\delta}\right).$$

Since  $\eta = 1$  on  $|x| \leq 1$  and  $\eta = 0$  on  $|x| \geq 2$ , this implies

$$2^{j-1}\delta \leq |x| \leq 2^{j+1}\delta.$$

Thus each  $\psi_j$  is supported in a dyadic annulus. For  $|x| > \delta$  we compute

$$\sum_{j=0}^N \psi_j(x) = \eta\left(\frac{x}{2^N\delta}\right) - \eta\left(\frac{x}{\delta}\right).$$

Since  $|x| > \delta$ , we have  $\eta(x/\delta) = 0$ . Moreover

$$\frac{x}{2^N \delta} \rightarrow 0$$

as  $N \rightarrow \infty$ , hence

$$\eta\left(\frac{x}{2^N \delta}\right) \rightarrow 1.$$

Therefore

$$\sum_{j \geq 0} \psi_j(x) = 1 \quad \text{for } |x| > \delta.$$

Let  $k \geq 1$ . Using the chain rule we obtain

$$\nabla^k \left( \eta\left(\frac{x}{2^j \delta}\right) \right) = (2^j \delta)^{-k} (\nabla^k \eta)\left(\frac{x}{2^j \delta}\right).$$

Since  $\eta \in C_c^\infty$ , its derivatives are bounded. Hence there exists  $C_k > 0$  such that

$$\left| \nabla^k \left( \eta\left(\frac{x}{2^j \delta}\right) \right) \right| \leq \frac{C_k}{(2^j \delta)^k}.$$

Differentiating the definition of  $\psi_j$  gives

$$\nabla^k \psi_j(x) = \nabla^k \left( \eta\left(\frac{x}{2^j \delta}\right) \right) - \nabla^k \left( \eta\left(\frac{x}{2^{j-1} \delta}\right) \right).$$

Therefore

$$|\nabla^k \psi_j(x)| \leq \frac{C_k}{(2^j \delta)^k} + \frac{C_k}{(2^{j-1} \delta)^k}.$$

Thus there exists  $C'_k > 0$  such that

$$|\nabla^k \psi_j(x)| \leq \frac{C'_k}{(2^j \delta)^k}.$$

On the support of  $\psi_j$  we have

$$|x| \sim 2^j \delta,$$

hence the derivative estimate can also be written as

$$|\nabla^k \psi_j(x)| \leq C_k |x|^{-k}.$$

□

**Lemma 3.17.** *Let  $f \in C^2(\mathbb{R}^n)$ . Then for  $0 < \alpha < 1$*

$$[f]_{C^\alpha} \lesssim \|\nabla f\|_{L^\infty}^{1-\alpha} \|\nabla^2 f\|_{L^\infty}^\alpha.$$

*Proof.* Let  $x, y \in \mathbb{R}^n$  and set  $h = |x - y|$ . Using the fundamental theorem of calculus along the segment  $x + t(y - x)$  we write

$$\nabla f(x) - \nabla f(y) = \int_0^1 \nabla^2 f(x + t(y - x))(y - x) dt.$$

Hence

$$|\nabla f(x) - \nabla f(y)| \leq |x - y| \|\nabla^2 f\|_{L^\infty}.$$

We also have the trivial bound

$$|\nabla f(x) - \nabla f(y)| \leq 2 \|\nabla f\|_{L^\infty}.$$

Combining the two inequalities we obtain

$$|\nabla f(x) - \nabla f(y)| \leq \min \left( 2 \|\nabla f\|_{L^\infty}, |x - y| \|\nabla^2 f\|_{L^\infty} \right).$$

A standard interpolation inequality for positive numbers gives

$$\min(A, B) \leq A^{1-\alpha} B^\alpha.$$

Applying this with

$$A = 2 \|\nabla f\|_{L^\infty}, \quad B = |x - y| \|\nabla^2 f\|_{L^\infty},$$

we obtain

$$|\nabla f(x) - \nabla f(y)| \leq C \|\nabla f\|_{L^\infty}^{1-\alpha} \|\nabla^2 f\|_{L^\infty}^\alpha |x - y|^\alpha.$$

Dividing by  $|x - y|^\alpha$  and taking the supremum over  $x \neq y$  yields the desired estimate.

□

**Proposition 3.18** ([2] Lemma 1). *If  $p > 0$  is sufficiently small, then for all sufficiently small  $\varepsilon > 0$  there exists  $F_\varepsilon : \mathbb{R}^{d-1} \rightarrow \mathbb{R}$ ,  $\text{supp}(F_\varepsilon) \subset D(0, \varepsilon^{1/2})$  such that:*

$$\int_{\mathbb{R}^{d-1}} \left( \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 \right) < -\eta, \quad (3.2)$$

*with  $\eta$  independent of  $\varepsilon$ . Moreover,  $|\nabla \hat{F}_\varepsilon| \lesssim \min(\varepsilon^{-d}, |x|^{-d})$  on  $\mathbb{R}^{d-1}$ . Where the hat notation  $\hat{F}_\varepsilon$ , denotes the harmonic extension of  $F_\varepsilon$  on the upper half semi-space  $\mathcal{H}^d$ .*

*Proof.* Take  $\delta = \varepsilon^{1/2}$  and for  $j \geq 0$ , let  $\psi_j$ , such as in Lemma 3.16. Consider also,  $G_\varepsilon$  such as in Lemma 3.15.

We set  $\rho_\varepsilon^j := \psi_j G_\varepsilon$  and define  $\psi := \sum_j \psi_j$ ,  $\rho_\varepsilon := \sum_j \rho_\varepsilon^j = \psi G_\varepsilon$ ,  $F_\varepsilon = G_\varepsilon - \rho_\varepsilon$ . Observe that  $\rho_\varepsilon^j, \rho_\varepsilon$  are functions defined on  $\mathbb{R}^{d-1}$ .

Using  $\nabla_T$  to denote the gradient with respect to the first  $d-1$  coordinates we claim that,  $|\nabla_T^k \rho_\varepsilon^j| \lesssim \varepsilon(2^j \delta)^{-(d+k)}$  for  $k = 0, 1, 2$ .

Indeed, we first notice that  $\rho_\varepsilon^j(x') = -\psi_j(x') \frac{\varepsilon}{(|x'|^2 + \varepsilon^2)^{d/2}}$ ,  $x' \in \mathbb{R}^{d-1}$ . For  $k = 0$ ,

$$|\rho_\varepsilon^j(x')| = |\psi_j(x') \frac{\varepsilon}{(|x'|^2 + \varepsilon^2)^{d/2}}| \leq \left| \frac{\varepsilon}{(|x'|^2 + \varepsilon^2)^{d/2}} \right| \leq \frac{\varepsilon}{|x'|^d} \lesssim \varepsilon(2^j \delta)^{-d}$$

whenever  $x' \in \text{supp}(\psi_j)$ , being zero otherwise. For  $k = 1$  we have

$$\nabla_T(\psi_j(x') \frac{-\varepsilon}{(|x'|^2 + \varepsilon^2)^{d/2}}) = \nabla_T(\psi_j(x')) \frac{-\varepsilon}{(|x'|^2 + \varepsilon^2)^{d/2}} + \psi_j(x') \nabla_T \left( \frac{-\varepsilon}{(|x'|^2 + \varepsilon^2)^{d/2}} \right)$$

To estimate  $\nabla_T(\frac{-\varepsilon}{(|x'|^2 + \varepsilon^2)^{d/2}})$ , we readily see that,

$$\partial_i \left( \frac{-\varepsilon}{(|x'|^2 + \varepsilon^2)^{d/2}} \right) = d\varepsilon(|x'|^2 + \varepsilon^2)^{-(d+2)/2} x_i.$$

Therefore, the euclidean norm for the gradient is

$$\left| \nabla_T \left( \frac{-\varepsilon}{(|x'|^2 + \varepsilon^2)^{d/2}} \right) \right| = d\varepsilon(|x'|^2 + \varepsilon^2)^{-(d+2)/2} |x'| \lesssim \varepsilon(2^j \delta)^{-(d+1)}.$$

Using the bound  $|\nabla \psi_j| \leq (2^j \delta)^{-1}$  from Lemma 3.16, combined with the  $k = 0$  case, and what we have just showed, we have  $|\nabla_T(\rho_\varepsilon^j)| \lesssim \varepsilon(2^j \delta)^{-(d+1)}$ .

Applying the product rule again,

$$\nabla_T^2(\psi_j G_\varepsilon) = (\nabla_T^2 \psi_j) G_\varepsilon + 2(\nabla_T \psi_j)(\nabla G_\varepsilon) + \psi_j \nabla_T^2 G_\varepsilon,$$

and performing analogous calculations as in the cases  $k = 0, 1$  we obtain what was claimed. Above we have abused the notation  $G_\varepsilon = G_\varepsilon(x', 0)$ .

Since,  $|\nabla_T^k \rho_\varepsilon^j| \lesssim \varepsilon(2^j \delta)^{-(d+k)}$  we have  $\|\nabla_T \rho_\varepsilon^j\|_{L^1} \lesssim \varepsilon(2^j \delta)^{-2}$  and  $\|\nabla_T \rho_\varepsilon^j\|_{C^\alpha} \lesssim \varepsilon(2^j \delta)^{-(d+1+\alpha)}$ .

Indeed, since  $\rho_\varepsilon^j$  is supported in the annulus

$$2^{j-1} \delta \leq |x'| \leq 2^{j+1} \delta,$$

its measure satisfies

$$|\text{supp}(\rho_\varepsilon^j)| \lesssim (2^j \delta)^{d-1}.$$

From the pointwise estimate

$$|\nabla_T \rho_\varepsilon^j(x')| \leq \varepsilon (2^j \delta)^{-(d+1)},$$

we obtain

$$\|\nabla_T \rho_\varepsilon^j\|_{L^1} = \int |\nabla_T \rho_\varepsilon^j(x')| dx'.$$

Using the bound and the support estimate,

$$\|\nabla_T \rho_\varepsilon^j\|_{L^1} \leq |\text{supp}(\rho_\varepsilon^j)| \sup |\nabla_T \rho_\varepsilon^j|.$$

Therefore

$$\|\nabla_T \rho_\varepsilon^j\|_{L^1} \lesssim (2^j \delta)^{d-1} \cdot \varepsilon (2^j \delta)^{-(d+1)}.$$

Simplifying the powers,

$$(2^j \delta)^{d-1-(d+1)} = (2^j \delta)^{-2}.$$

Thus

$$\|\nabla_T \rho_\varepsilon^j\|_{L^1} \lesssim \varepsilon (2^j \delta)^{-2}.$$

Since,  $\|\nabla_T \rho_\varepsilon^j\|_{L^1} \leq \varepsilon (2^j \delta)^{-2}$ , we use Theorem 3.9, i.e.,  $\partial_n \hat{\rho}_\varepsilon^j = \sum R_k(\partial_k \rho_\varepsilon^j)$ , and apply the weak-type  $(1, 1)$  inequality, Theorem 3.13, to obtain:

$$\left| \left\{ x \in \mathbb{R}^{d-1} : \left| \frac{\partial \hat{\rho}_\varepsilon^j}{\partial n}(x) \right| > \lambda \right\} \right| \leq C \lambda^{-1} \|\nabla_T \rho_\varepsilon^j\|_{L^1} \lesssim \varepsilon (2^j \delta)^{-2} \lambda^{-1}.$$

We now estimate the  $C^\alpha$  norm of  $\nabla_T \rho_\varepsilon^j$ . We use the interpolation inequality from Lemma 3.17

Applying this with  $f = \rho_\varepsilon^j$  gives

$$\|\nabla_T \rho_\varepsilon^j\|_{C^\alpha} \lesssim \|\nabla_T \rho_\varepsilon^j\|_{L^\infty}^{1-\alpha} \|\nabla_T^2 \rho_\varepsilon^j\|_{L^\infty}^\alpha.$$

Using the pointwise bounds

$$\|\nabla_T \rho_\varepsilon^j\|_{L^\infty} \lesssim \varepsilon (2^j \delta)^{-(d+1)},$$

$$\|\nabla_T^2 \rho_\varepsilon^j\|_{L^\infty} \lesssim \varepsilon (2^j \delta)^{-(d+2)},$$

we obtain

$$\|\nabla_T \rho_\varepsilon^j\|_{C^\alpha} \lesssim (\varepsilon(2^j \delta)^{-(d+1)})^{1-\alpha} (\varepsilon(2^j \delta)^{-(d+2)})^\alpha.$$

Simplifying the powers of  $\varepsilon$ ,

$$\varepsilon^{1-\alpha+\alpha} = \varepsilon.$$

For the powers of  $(2^j \delta)$  we compute

$$(d+1)(1-\alpha) + (d+2)\alpha = d+1+\alpha.$$

Hence

$$\|\nabla_T \rho_\varepsilon^j\|_{C^\alpha} \lesssim \varepsilon(2^j \delta)^{-(d+1+\alpha)}.$$

Moreover, from  $\|\nabla_T \rho_\varepsilon^j\|_{C^\alpha} \lesssim \varepsilon(2^j \delta)^{-(d+1+\alpha)}$  we can establish the bound  $[\partial_n \hat{\rho}_\varepsilon^j]_{C^\alpha} \lesssim \varepsilon(2^j \delta)^{-(d+1+\alpha)}$ .

In fact, Theorem 3.11, give us  $[R(\nabla_T \rho_\varepsilon^j)]_{C^\alpha} \lesssim \|\nabla_T \rho_\varepsilon^j\|_{C^\alpha} \lesssim \varepsilon(2^j \delta)^{-(d+1+\alpha)}$ . Applying Theorem 3.9 again, we get  $[\partial_n \hat{\rho}_\varepsilon^j]_{C^\alpha} \lesssim \varepsilon(2^j \delta)^{-(d+1+\alpha)}$ .

In possession of both inequalities:

$$\left| \left\{ x \in \mathbb{R}^{d-1} : \left| \frac{\partial \hat{\rho}_\varepsilon^j}{\partial n}(x) \right| > \lambda \right\} \right| \leq C\lambda^{-1} \|\nabla_T \rho_\varepsilon^j\|_{L^1} \leq C_W \varepsilon(2^j \delta)^{-2} \lambda^{-1}; \quad (3.3)$$

$$\left[ \frac{\partial \hat{\rho}_\varepsilon^j}{\partial n} \right]_{C^\alpha} \leq C_H \varepsilon(2^j \delta)^{-(d+1+\alpha)}; \quad (3.4)$$

To obtain a point-wise estimate for  $\partial_n \hat{\rho}_\varepsilon^j$ , we define the region  $B_j = \{x' \in \mathbb{R}^{d-1} : |x'| < 2^{j+2} \delta\}$ . The Lebesgue measure of this region is  $|B_j| = v_{d-1}(2^{j+2} \delta)^{d-1}$ , where  $v_{d-1}$  denotes the volume of the unit ball in  $\mathbb{R}^{d-1}$ .

We choose the value of  $\lambda$  such that the measure of the set where  $|\partial_n \hat{\rho}_\varepsilon^j|$  exceeds  $\lambda$  is exactly half the measure of  $B_j$ :

$$\frac{C_W \varepsilon(2^j \delta)^{-2}}{\lambda} = \frac{1}{2} v_{d-1} (2^{j+2} \delta)^{d-1}.$$

Solving for  $\lambda$ , we obtain the value:

$$\lambda = \frac{2C_W \varepsilon(2^j \delta)^{-2}}{v_{d-1} 2^{2(d-1)} (2^j \delta)^{d-1}} = \left( \frac{2C_W}{v_{d-1} 2^{2d-2}} \right) \varepsilon(2^j \delta)^{-(d+1)}.$$

Since  $|\{x' \in B_j : |\partial_n \hat{\rho}_\varepsilon^j| > \lambda\}| < |B_j|$ , there must exist a point  $x'_0 \in B_j$  such that  $|\partial_n \hat{\rho}_\varepsilon^j(x'_0)| \leq \lambda$ . Now, for any  $x' \in B_j$ , we apply the Holder inequality with respect to this point  $x'_0$ :

$$\begin{aligned} |\partial_n \hat{\rho}_\varepsilon^j(x')| &\leq |\partial_n \hat{\rho}_\varepsilon^j(x'_0)| + |\partial_n \hat{\rho}_\varepsilon^j(x') - \partial_n \hat{\rho}_\varepsilon^j(x'_0)| \\ &\leq \lambda + C_H \varepsilon (2^j \delta)^{-(d+1+\alpha)} |x' - x'_0|^\alpha. \end{aligned}$$

As  $x', x'_0 \in B_j$ , the maximum distance  $|x' - x'_0|$  is the diameter of the ball,  $2^{j+3}\delta$ . Thus:

$$\begin{aligned} |\partial_n \hat{\rho}_\varepsilon^j(x')| &\leq \left( \frac{2C_W}{v_{d-1} 2^{2d-2}} \right) \varepsilon (2^j \delta)^{-(d+1)} + C_H \varepsilon (2^j \delta)^{-(d+1+\alpha)} (2^3 \cdot 2^j \delta)^\alpha \\ &= \left( \frac{2C_W}{v_{d-1} 4^{d-1}} + 8^\alpha C_H \right) \varepsilon (2^j \delta)^{-(d+1)}. \end{aligned}$$

Therefore, for  $x' \in B_j$ , we obtain the desired pointwise estimate:

$$\left| \frac{\partial \hat{\rho}_\varepsilon^j}{\partial n}(x') \right| \lesssim \varepsilon (2^j \delta)^{-(d+1)}, \quad |x'| < 2 \cdot 2^{j+1} \delta. \quad (3.5)$$

Moreover, for  $|x'| \geq 2 \cdot 2^{j+1} \delta$  we have

$$\begin{aligned} \partial_n \hat{\rho}_\varepsilon^j(x', 0) &= -\partial_{x_d} \big|_{x_d=0} \int_{\mathbb{R}^{d-1}} P(x' - y', x_d) \rho_\varepsilon^j(y') dy' \\ (\text{by Dominated Convergence}) &= -c_d \int_{|y'| < 2^{j+1} \delta} \partial_{x_d} \frac{x_d}{(|x' - y'|^2 + x_d^2)^{d/2}} \rho_\varepsilon^j(y') dy' \\ (x_d = 0) &= -c_d \int_{|y'| < 2^{j+1} \delta} |x' - y'|^{-d} \rho_\varepsilon^j(y') dy'. \end{aligned}$$

We observe that, since  $|x'| \geq 2 \cdot 2^{j+1}$  and  $|y'| < 2^{j+1} < \frac{1}{2}|x'|$ , reverse triangle inequality gives  $|x' - y'| \geq |x'| - |y'| > \frac{1}{2}|x'|$ . By taking norm on the previous integral and using that  $|\rho_\varepsilon^j| \lesssim \varepsilon (2^j \delta)^{-d}$  as we already proved, together with  $|x' - y'|^{-d} \lesssim |x'|^{-d}$  we obtain

$$|\partial_n \hat{\rho}_\varepsilon^j(x', 0)| \lesssim \varepsilon (2^j \delta)^{-d} |x'|^{-d} (2^j \delta)^{d-1} = \varepsilon (2^j \delta)^{-1} |x'|^{-d}.$$

From now on, we denote the points of  $\mathbb{R}^{d-1}$  simply by  $x$  when there is no danger of misconception. We then obtain  $|\partial_n \hat{\rho}_\varepsilon^j| \lesssim \varepsilon (2^j \delta)^{-1} \min\{|x|^{-d}, (2^j \delta)^{-d}\}$ . Summing over  $j$  we get

$$|\partial_n \hat{\rho}_\varepsilon| \lesssim \varepsilon \delta^{-1} \min\{\delta^{-d}, |x|^{-d}\}. \quad (3.6)$$

For  $p$  small enough, i.e.,  $p < \frac{d-1}{d}$  we can take  $\eta > 0$  such as:

$$\begin{aligned} \int_{\mathbb{R}^{d-1}} \left( \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 \right) &= \int_{\mathbb{R}^{d-1}} \left( \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p - 1 \right) + \\ &\quad + \int_{\mathbb{R}^{d-1}} \left( \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right) \\ &\leq -2\eta + \int_{|x|<1} \left( \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right) + \\ &\quad + \int_{|x|>1} \left( \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right), \end{aligned}$$

by Lemma 3.15. Recall that the function  $F_\varepsilon$  is constructed as a modification of  $G_\varepsilon$  such that in  $x_d = 0$ :

$$\frac{d\hat{F}_\varepsilon}{dn} = \frac{dG_\varepsilon}{dn} - \frac{d\hat{\rho}_\varepsilon}{dn}.$$

We then rewrite the integrand of

$$\int_{|x|<1} \left| \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right| dx$$

as:

$$\left| \left| 1 + \frac{dG_\varepsilon}{dn} - \frac{d\hat{\rho}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right|$$

For  $0 < p < 1$ , it holds that  $(a+b)^p \leq a^p + b^p$ . This implies the general inequality for the modulus of the difference of powers:

$$||X - Y|^p - |X|^p| \leq |Y|^p$$

By setting  $X = 1 + \frac{dG_\varepsilon}{dn}$  and  $Y = \frac{d\hat{\rho}_\varepsilon}{dn}$ , we obtain:

$$\left| \left| 1 + \frac{dG_\varepsilon}{dn} - \frac{d\hat{\rho}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right| \leq \left| \frac{d\hat{\rho}_\varepsilon}{dn} \right|^p$$

Integrating both sides and equation (3.6) we get:

$$\int_{|x|<1} \left| \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right| dx \leq \int_{|x|\leq 1} \left| \frac{d\hat{\rho}_\varepsilon}{dn} \right|^p dx \lesssim \varepsilon \delta^{-1} \min\{\delta^{-d}, 1\} \leq \varepsilon^{1/2}.$$

That goes to zero as  $\varepsilon \rightarrow 0$ . Outside the unitary ball from the definition of  $G_\varepsilon$ , we have the bound:

$$\left| \frac{dG_\varepsilon}{dn} \right| \leq \varepsilon |x|^{-d}$$

as computed previously.

For  $|x| > 1$ , as  $\varepsilon \rightarrow 0$ , it follows that  $\left| \frac{dG_\varepsilon}{dn} \right| \rightarrow 0$ . Consequently, the term  $1 + \frac{dG_\varepsilon}{dn}$  approaches 1.

Specifically, for sufficiently small  $\varepsilon$  (say  $\varepsilon < 1/4$ ), we have  $1 + \frac{dG_\varepsilon}{dn} \in [3/4, 5/4]$ . Since equation (3.6) assures that  $\frac{d\hat{\rho}_\varepsilon}{dn}$  vanishes uniformly on  $|x| > 1$ , we can choose  $\varepsilon$  small enough such that  $\left| \frac{d\hat{\rho}_\varepsilon}{dn} \right| < 1/4$ .

Let  $h(u) = |u|^p$ . For  $p > 0$ ,  $h$  is differentiable for  $u \neq 0$ . Given the bounds above, the values

$$u_1 = 1 + \frac{d\hat{F}_\varepsilon}{dn} \quad \text{and} \quad u_2 = 1 + \frac{dG_\varepsilon}{dn}$$

lie in a compact interval  $[1/2, 3/2]$ , which is strictly contained in  $(0, \infty)$ .

By the Mean Value Theorem, there exists an intermediate value  $\xi \in [u_1, u_2] \subset [1/2, 3/2]$  such that:

$$\left| \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right| = |h'(\xi)| \cdot \left| \frac{d\hat{\rho}_\varepsilon}{dn} \right|$$

where  $h'(\xi) = p\xi^{p-1}$ . Because  $\xi$  is trapped in  $[1/2, 3/2]$ , the derivative is bounded by its value at the left endpoint (since  $p - 1 < 0$ ):

$$|h'(\xi)| \leq p(1/2)^{p-1} = C_p.$$

This yields the point-wise inequality:

$$\left| \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right| \leq C_p \left| \frac{d\hat{\rho}_\varepsilon}{dn} \right|.$$

Integrating the previous result over  $|x| > 1$ :

$$\int_{|x|>1} \left| \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{dG_\varepsilon}{dn} \right|^p \right| dx \leq C_p \int_{|x|>1} \left| \frac{d\hat{\rho}_\varepsilon}{dn} \right| dx.$$

Substituting the bound for  $\rho_\varepsilon$  and evaluating the integral:

$$C_p \int_{|x|>1} \varepsilon \delta^{-1} |x|^{-d} dx \leq C_p \varepsilon \delta^{-1} \int_1^\infty r^{-d} r^{d-2} dr = C_p \varepsilon \delta^{-1},$$

that also goes to zero as  $\varepsilon \rightarrow 0$ . That finally proves

$$\int \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 < -\eta.$$

To prove  $|\nabla \hat{F}_\varepsilon| \lesssim \min\{\varepsilon^{-d}, |x|^{-d}\}$  on  $\mathbb{R}^{d-1}$  we write  $|\nabla \hat{F}_\varepsilon| \leq |\nabla G_\varepsilon| + |\nabla \hat{\rho}_\varepsilon|$  and compute each part.

Denoting,  $r = \sqrt{\sum_{i=1}^{d-1} x_i^2 + (x_d + \varepsilon)^2}$  we have the tangential derivatives:

$$\frac{\partial G_\varepsilon}{\partial x_i} = \frac{d(x_d + \varepsilon)x_i}{r^{d+2}},$$

and the normal derivative:

$$\frac{\partial G_\varepsilon}{\partial x_d} = -\frac{1}{r^d} + \frac{d(x_d + \varepsilon)^2}{r^{d+2}}.$$

Then, using  $x_d = 0$ :

$$|\nabla G_\varepsilon| = \left( \sum_{i=1}^{d-1} \left( \frac{d\varepsilon x_i}{r^{d+2}} \right)^2 + \left( \frac{(d-1)\varepsilon^2 - |x|^2}{r^{d+2}} \right)^2 \right)^{1/2}.$$

Simplify the numerator

$$|\nabla G_\varepsilon| = \frac{\sqrt{d^2\varepsilon^2|x|^2 + (d-1)^2\varepsilon^4 - 2(d-1)\varepsilon^2|x|^2 + |x|^4}}{r^{d+2}}.$$

Since the numerator is  $O(r^2)$  and  $r = \sqrt{|x|^2 + \varepsilon^2}$ :

$$|\nabla G_\varepsilon| \leq \frac{C \cdot r^2}{r^{d+2}} = \frac{C}{r^d},$$

From  $|\nabla G_\varepsilon| \leq \frac{C}{(|x|^2 + \varepsilon^2)^{d/2}}$  we get  $|\nabla G_\varepsilon| \lesssim \min(\varepsilon^{-d}, |x|^{-d})$ .

To  $\nabla \hat{\rho}_\varepsilon^j$  recall equation (3.6):

$$|\partial_{x_d} \hat{\rho}_\varepsilon^j| = |\partial_n \hat{\rho}_\varepsilon| \lesssim \varepsilon \delta^{-1} \min\{\delta^{-d}, |x|^{-d}\} \leq \min\{\varepsilon^{-d}, |x|^{-d}\} \quad \text{for } \varepsilon < 1.$$

Whereas for the first  $d-1$  coordinates the bound can be obtained by direct computation, as we did for  $G_\varepsilon$  and in the beginning of the proof. Finally getting  $|\nabla \hat{F}_\varepsilon| \lesssim \min\{\varepsilon^{-d}, |x|^{-d}\}$ .

□

**Proposition 3.19** ([2] Lemma 2). *If  $Q \in \mathbb{R}^{d-1}$  is a cube with center  $a_Q$  and  $I : Q \rightarrow \mathbb{R}$  is a function such that  $N^{d-1}|I(a_Q)|^{-1} \sup_{x \in Q} |I(x) - I(a_Q)|$  is sufficiently small, not depending on  $N, \varepsilon$ , then for  $N$  large enough there is a positive constant  $\beta(N)$  such that:*

$$\left( \int_Q \left| I(x) + \frac{d\hat{F}_\varepsilon}{dn}(N \cdot \ell(Q)^{-1}(x - a_Q)) \right|^p dx \right)^{1/p} < e^{-2\beta} |I(a_Q)| |Q|^{1/p}.$$

*Proof.* We define a transformation to map the generic cube  $Q$  to the standard cube  $Q(N) = [-\frac{N}{2}, \frac{N}{2}]^{d-1}$ . Let:

$$y = N\ell(Q)^{-1}(x - a_Q) \implies x = a_Q + \frac{\ell(Q)}{N}y.$$

The Jacobian determinant of this transformation is:

$$dx = \left( \frac{\ell(Q)}{N} \right)^{d-1} dy.$$

As  $x$  ranges over  $Q$ ,  $y$  ranges over  $Q(N)$ . Substituting into the integral:

$$\int_{Q(N)} \left| I\left(a_Q + \frac{\ell(Q)}{N}y\right) + \frac{d\hat{F}_\varepsilon}{dn}(y) \right|^p \left( \frac{\ell(Q)}{N} \right)^{d-1} dy. \quad (3.7)$$

Assume  $I(a_Q) \neq 0$ . We normalize the expression by dividing by  $|I(a_Q)|^p$  inside the integral. Define the normalized function  $I_{norm}(y)$  as:

$$I_{norm}(y) = \frac{I\left(a_Q + \frac{\ell(Q)}{N}y\right)}{I(a_Q)}.$$

Note that  $I_{norm}(0) = 1$ . The inequality becomes:

$$|I(a_Q)| \left( \frac{\ell(Q)}{N} \right)^{\frac{d-1}{p}} \left( \int_{Q(N)} \left| I_{norm}(y) + \frac{d\hat{F}_\varepsilon}{dn}(y) \right|^p dy \right)^{1/p} < e^{-2\beta} |I(a_Q)| \ell(Q)^{\frac{d-1}{p}}.$$

Canceling  $|I(a_Q)|$  and  $\ell(Q)^{(d-1)/p}$  from both sides, we obtain:

$$N^{-\frac{d-1}{p}} \left( \int_{Q(N)} \left| I_{norm}(y) + \frac{d\hat{F}_\varepsilon}{dn}(y) \right|^p dy \right)^{1/p} < e^{-2\beta}.$$

Raising both sides to the power  $p$ :

$$\frac{1}{N^{d-1}} \int_{Q(N)} \left| I_{norm}(y) + \frac{d\hat{F}_\varepsilon}{dn}(y) \right|^p dy < e^{-2\beta p}.$$

Since  $e^{-2\beta p} < 1$  for  $\beta > 0$  let us call this  $1 - \alpha$  with  $\alpha > 0$ , that is

$$\int_{Q(N)} \left| I_{norm}(y) + \frac{d\hat{F}_\varepsilon}{dn}(y) \right|^p dy < (1 - \alpha)N^{d-1},$$

which, recalling that  $\int_{Q(N)} 1 dy = N^{d-1}$ , is equivalent to

$$\int_{Q(N)} \left( \left| I_{norm}(y) + \frac{d\hat{F}_\varepsilon}{dn}(y) \right|^p - 1 \right) dy < -\alpha N^{d-1}.$$

Thus, the result reduces to showing there exists  $\alpha N^{d-1} = \eta > 0$  such that:

$$\int_{Q(N)} \left( \left| I_{norm}(y) + \frac{d\hat{F}_\varepsilon}{dn}(y) \right|^p - 1 \right) dy \leq -\eta$$

provided the oscillation term  $N^{d-1} \|I_{norm} - 1\|_\infty$  is sufficiently small. We simplify the notation and write  $I = I_{norm}$ .

We decompose the integral into two terms by adding and subtracting the case where the function is exactly 1:

$$\begin{aligned} \int_{Q(N)} \left( \left| I + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 \right) dy &= \int_{Q(N)} \left( \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 \right) dy \\ &\quad + \underbrace{\int_{Q(N)} \left( \left| I + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p \right) dy}_{\text{Error Term } (\Delta)}. \end{aligned}$$

We split the domain  $Q(N)$  into  $D(0, 1)$  and  $Q(N) \setminus D(0, 1)$ . In this region, the term  $1 + \frac{d\hat{F}_\varepsilon}{dn}$  is bounded away from zero because the correction  $\frac{d\hat{F}_\varepsilon}{dn}$  decays far from the origin, as we argue in the proof of Proposition 3.18.

Let  $f(t) = |t|^p$ . Since  $f$  is smooth away from the origin, we apply the Mean Value Theorem:

$$||I + A|^p - |1 + A|^p| \leq \sup_{\xi \in [I, 1]} p|\xi + A|^{p-1}|I - 1| \leq C|I - 1|$$

where  $A = \frac{d\hat{F}_\varepsilon}{dn}$ . Integrating over the outer region:

$$\int_{Q(N) \setminus D(0, 1)} C|I(y) - 1| dy \leq C\|I - 1\|_\infty \cdot \text{Vol}(Q(N)) \leq C\|I - 1\|_\infty N^{d-1}.$$

Within the unit disk, the term  $1 + \frac{d\hat{F}_\varepsilon}{dn}$  might be small or zero. We use the inequality  $|a + b|^p \leq |a|^p + |b|^p$  for  $0 < p < 1$ :

$$||I + A|^p - |1 + A|^p| \leq |I - 1|^p.$$

Integrating over the disk  $D(0, 1)$ , whose volume is independent of  $N$ :

$$\int_{D(0,1)} |I - 1|^p dy \leq C \|I - 1\|_\infty^p.$$

For  $\|I - 1\|_\infty < 1$ , we have  $\|I - 1\|_\infty^p \leq C' \|I - 1\|_\infty$ .

From Proposition 3.18, we know that for the global integral:

$$\int_{\mathbb{R}^{d-1}} \left( \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 \right) dy < -2\eta.$$

By choosing  $N$  large enough, the integral over  $Q(N)$  captures the majority of this negative value:

$$\int_{Q(N)} \left( \left| 1 + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 \right) dy \leq -\frac{3}{2}\eta.$$

Combining the estimates:

$$\int_{Q(N)} \left( \left| I + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 \right) dy \leq -1.5\eta + C \|I - 1\|_\infty N^{d-1} + C \|I - 1\|_\infty.$$

By the hypothesis,  $N^{d-1} \|I - 1\|_\infty$  is sufficiently small (say,  $C \|I - 1\|_\infty N^{d-1} + C \|I - 1\|_\infty \leq 0.5\eta$ ). Thus:

$$\int_{Q(N)} \left( \left| I + \frac{d\hat{F}_\varepsilon}{dn} \right|^p - 1 \right) dy \leq -1.5\eta + 0.5\eta = -\eta.$$

□

### 3.2.2 The Inductive Algorithm

The construction is recursive and based upon a stopping time algorithm, just as in Calderon-Zygmund decomposition argument.

We start with  $u_0 \in C^\infty(\mathbb{R}^{d-1})$  vanishing on  $Q(1) = [-1/2, 1/2]^{d-1}$ , and large enough  $N$ , to be determined later on. Also taking  $\beta = \beta(N)$  such as Proposition 3.19 holds, we chose sequences  $K_n \rightarrow \infty$  and  $\varepsilon_n \searrow 0$  such as

$$\sum \varepsilon_n^{-d} K_{n+1} e^{-\beta n} < \infty \quad (3.8)$$

$$\sum K_{n+1}^{-p} + \varepsilon_{n+1}^{(d-1)/2} \text{ is sufficiently small.} \quad (3.9)$$

We verify that such sequences do exist.

**Lemma 3.20.** *By taking  $\varepsilon_n = C^{-1}n^{-2}$  and  $K_n = Cn^{2/p}$  where  $C > 0$  is a constant to be chosen later, we have that  $K_n \rightarrow \infty$  and  $\varepsilon_n \searrow 0$  and equations (3.8) and (3.9) holds.*

*Proof.* We have trivially that,  $K_n \rightarrow \infty$  and  $\varepsilon_n \searrow 0$ . We begin by checking that equation (3.8) holds. Observe that

$$\varepsilon_n^{-d} = (C^{-1}n^{-2})^{-d} = C^d n^{2d},$$

and

$$K_{n+1} = C(n+1)^{2/p}.$$

Hence

$$\varepsilon_n^{-d} K_{n+1} e^{-\beta n} = C^{d+1} n^{2d} (n+1)^{2/p} e^{-\beta n}.$$

Since  $(n+1)^{2/p} \leq 2^{2/p} n^{2/p}$  for  $n \geq 1$ , it follows that

$$\varepsilon_n^{-d} K_{n+1} e^{-\beta n} \leq C' n^{2d+2/p} e^{-\beta n}.$$

Because exponential decay dominates any polynomial growth, we have

$$\sum_{n=1}^{\infty} n^{2d+2/p} e^{-\beta n} < \infty,$$

and therefore

$$\sum_{n=1}^{\infty} \varepsilon_n^{-d} K_{n+1} e^{-\beta n} < \infty.$$

Next we verify equation (3.9). We compute

$$K_{n+1}^{-p} = (C(n+1)^{2/p})^{-p} = C^{-p} (n+1)^{-2},$$

and

$$\varepsilon_{n+1}^{(d-1)/2} = (C^{-1}(n+1)^{-2})^{(d-1)/2} = C^{-(d-1)/2} (n+1)^{-(d-1)}.$$

Thus

$$\sum_{n=1}^{\infty} K_{n+1}^{-p} = C^{-p} \sum_{n=1}^{\infty} (n+1)^{-2} < \infty$$

and

$$\sum_{n=1}^{\infty} \varepsilon_{n+1}^{(d-1)/2} = C^{-(d-1)/2} \sum_{n=1}^{\infty} (n+1)^{-(d-1)} < \infty$$

(the latter series converges whenever  $d \geq 3$ ).

Finally, both sums in equation (3.9) are multiplied by negative powers of  $C$ . Hence, by choosing  $C$  sufficiently large, the quantity

$$\sum_{n=1}^{\infty} \left( K_{n+1}^{-p} + \varepsilon_{n+1}^{(d-1)/2} \right),$$

can be made arbitrarily small. This proves that the two conditions are compatible.  $\square$

The reason why we take the sequences  $\{\varepsilon_n\}, \{K_n\}$  satisfying equations (3.8) and (3.9) will become clear later on, for the time being we need only to bear in mind that  $K_n \rightarrow \infty$  and  $\varepsilon_n \searrow 0$ .

The goal of the first step is to modify an initial harmonic function  $u_0$  to a new function  $u_1$  such that the normal derivative  $\frac{du_1}{dn}$  becomes significantly smaller on a large portion of the boundary  $Q(1) \subset \mathbb{R}^{d-1}$  without disturbing too much elsewhere.

We have a constant  $A$  such that:

$$\left( \int_{V_0} \left| \frac{du_0}{dn} \right|^p \right)^{1/p} \leq A e^{-\beta \cdot 0} = A$$

where  $V_0 = Q(1)$  and  $A$  is going to very large, independent of  $n$ .

To move to Step 1, we choose a very small side length  $\delta_1$  (to be determined) such that  $\delta_1^{-1} \in \mathbb{Z}$ . We tile  $Q(1)$  with tiny cubes  $\mathcal{H}_1$  (the set of cubes of sides with length  $\delta_1$  whose union is  $Q(1)$ ) of side length  $\delta_1$ .

We define the "good" set  $\mathcal{G}_1$  by selecting only those tiny cubes  $Q \in \mathcal{H}_1$  where the  $L^p$  norm of the current normal derivative is not too large:

$$\mathcal{G}_1 = \left\{ Q \in \mathcal{H}_1 : \left( \frac{1}{|Q|} \int_Q \left| \frac{du_0}{dn} \right|^p \right)^{1/p} < K_1 \right\}$$

where  $K_1$  is a large constant chosen to ensure  $\mathcal{G}_1$  covers most of  $Q(1)$ .

For every cube  $Q \in \mathcal{G}_1$ , let  $a_Q$  be its center. We define the updated function  $u_1$  by adding a sum of localized "correction spikes":

$$u_1(x) = u_0(x) + \sum_{Q \in \mathcal{G}_1} \frac{du_0}{dn}(a_Q) F_{\varepsilon_1} \left( N \delta_1^{-1} (x - a_Q) \right) \frac{\delta_1}{N}$$

When we compute the normal derivative  $\frac{du_1}{dn}$  at a point  $x$  inside a specific cube  $Q$ , the derivative of the  $F_{\varepsilon_1}$  term is designed to be approximately  $-1$  (relative to the coefficient  $\frac{du_0}{dn}(a_Q)$ ).

$\frac{du_1}{dn}(x) \approx \frac{du_0}{dn}(x) - \frac{du_0}{dn}(a_Q)$ . Because  $u_0$  is smooth and  $\delta_1$  is extremely small,  $\frac{du_0}{dn}(x) \approx \frac{du_0}{dn}(a_Q)$ , so the two terms nearly cancel out.

The function  $F_{\varepsilon_1}$  has very small support (radius  $\approx \delta_1 \varepsilon_1^{1/2}$ ). This ensures that  $u_1$  remains equal to  $u_0$  (which is 0) on a large portion of the boundary.

By carefully choosing  $N$  and  $\delta_1$ , Proposition 3.24 guarantees that the new "energy" of the normal derivative is reduced by a factor of  $e^{-\beta}$ :

$$\left( \int_{V_1} \left| \frac{du_1}{dn} \right|^p \right)^{1/p} \leq A e^{-\beta}$$

where  $V_1$  is the union of the cubes in  $\mathcal{G}_1$ .

Formally the construction is the following, suppose we achieved the  $n$ -th stage; at this moment we have in hands:

1. A small positive number  $\delta_n$  such that  $\delta_n^{-1} \in \mathbb{Z}$ ;
2. A subset  $\mathcal{G}_n \subset \mathcal{H}_n$  where  $\mathcal{H}_n$  is the set of cubes of side  $\delta_n$  whose union is  $Q(1)$ ;
3. A smooth function  $u_n$  such that

$$\left( \int_{V_n} \left| \frac{d\hat{u}_n}{dn} \right|^p \right)^{1/p} \leq A e^{-\beta n},$$

for  $A$  large, independent of  $n$ , and  $V_n = \bigcup \{Q : Q \in \mathcal{G}_n\}$ .

Where we have started with  $\delta_0 = 1$  and  $\mathcal{G}_1 = \{Q(1)\}$ .

To do stage  $n+1$  choose a small  $\delta_{n+1}$ , as in Corollary 3.22, such that  $\delta_n/\delta_{n+1} \in \mathbb{Z}$ . Define  $\mathcal{G}_{n+1} \subset \mathcal{H}_{n+1}$  to be the set of cubes  $Q$  such that:

1. The cube  $Q' \in \mathcal{H}_n$  such as  $Q \subset Q'$  satisfies  $Q' \in \mathcal{G}_n$ ;
- 2.

$$\left( \frac{1}{|Q|} \int_Q \left| \frac{d\hat{u}_n}{dn} \right|^p \right)^{1/p} < K_{n+1} e^{-\beta n}.$$

From now on we drop the hat  $\hat{u}$  notation and represent simply by  $u$ , the harmonic extension from the boundary function. Define

$$u_{n+1}(x) = u_n(x) + \sum_{Q \in \mathcal{G}_{n+1}} \frac{du_n}{dn}(a_Q) F_{\varepsilon_{n+1}}(N \delta_{n+1}^{-1}(x - a_Q)) \frac{\delta_{n+1}}{N}$$

so that

$$\nabla u_{n+1}(x) = \nabla u_n(x) + \sum_{Q \in \mathcal{G}_{n+1}} \frac{du_n}{dn}(a_Q) \nabla F_{\varepsilon_{n+1}}(N \delta_{n+1}^{-1}(x - a_Q)).$$

**Proposition 3.21** ([2] Lemma 3).

$$\sum_{Q \in \mathcal{G}_{n+1}: |x - a_Q| > \rho} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q))| \leq CN^{-d}\delta_{n+1}\rho^{-1},$$

for all  $\rho > c \cdot \delta_{n+1}$  for some constant  $c$ , and  $x \in \mathbb{R}^{d-1}$ . With  $C$  depending on a lower bound for  $\rho/\delta_{n+1}$ .

*Proof.* By the last part of Proposition 3.18 we have, for every  $y \neq 0$ ,

$$|\nabla F_{\varepsilon}(y)| \leq C \min(\varepsilon^{-d}, |y|^{-d}).$$

We apply this estimate with

$$y = N\delta_{n+1}^{-1}(x - a_Q).$$

Since the sum in the statement is taken over those  $Q$  satisfying

$$|x - a_Q| > \rho,$$

we obtain

$$|y| = N\delta_{n+1}^{-1}|x - a_Q| > N\delta_{n+1}^{-1}\rho.$$

The parameters in the construction satisfy

$$\rho \geq c\delta_{n+1}$$

for some fixed constant  $c > 0$ . Hence

$$|y| > cN.$$

In particular  $|y|$  is bounded below by a positive constant independent of  $\varepsilon_{n+1}$ . Since  $\varepsilon_{n+1} \rightarrow 0$  in the construction, we have

$$|y| > \varepsilon_{n+1}.$$

Therefore the minimum in the estimate of Proposition 3.18 is realized by the second term, and we obtain

$$|\nabla F_{\varepsilon_{n+1}}(y)| \leq C|y|^{-d}.$$

Substituting  $y = N\delta_{n+1}^{-1}(x - a_Q)$  gives

$$|\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q))| \leq C(N\delta_{n+1}^{-1}|x - a_Q|)^{-d} = CN^{-d}\delta_{n+1}^d|x - a_Q|^{-d}.$$

Summing over all  $Q$  such that  $|x - a_Q| > \rho$  we obtain

$$\sum_{|x-a_Q|>\rho} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q))| \leq CN^{-d}\delta_{n+1}^d \sum_{|x-a_Q|>\rho} |x - a_Q|^{-d}.$$

It remains to estimate the discrete sum

$$S := \sum_{|x-a_Q|>\rho} |x - a_Q|^{-d}.$$

The points  $a_Q$  form a lattice in  $\mathbb{R}^{d-1}$  with spacing  $\delta_{n+1}$ . More precisely, each  $a_Q$  is the center of a cube

$$C_Q = a_Q + \left[ -\frac{\delta_{n+1}}{2}, \frac{\delta_{n+1}}{2} \right]^{d-1}$$

and the cubes  $\{C_Q\}$  are disjoint and tile  $\mathbb{R}^{d-1}$ . Moreover

$$|C_Q| = \delta_{n+1}^{d-1}.$$

For  $y \in C_Q$  we have

$$|x - y| \leq |x - a_Q| + |a_Q - y| \leq |x - a_Q| + C\delta_{n+1},$$

and

$$|x - y| \geq |x - a_Q| - C\delta_{n+1}.$$

Since  $\rho \geq c\delta_{n+1}$ , these inequalities imply that for  $|x - a_Q| > \rho$  and  $y \in C_Q$  one has

$$|x - y| \sim |x - a_Q|$$

with constants depending only on a lower bound for  $\rho/\delta_{n+1}$ .

Because the function  $r \mapsto r^{-d}$  is decreasing, this gives

$$|x - a_Q|^{-d} \leq C|x - y|^{-d} \quad (y \in C_Q).$$

Integrating over  $C_Q$  yields

$$\delta_{n+1}^{d-1}|x - a_Q|^{-d} = \int_{C_Q} |x - a_Q|^{-d} dy \leq C \int_{C_Q} |x - y|^{-d} dy.$$

Summing over all  $Q$  with  $|x - a_Q| > \rho$  we obtain

$$\delta_{n+1}^{d-1} \sum_{|x-a_Q|>\rho} |x - a_Q|^{-d} \leq C \int_{|x-y|>\rho/2} |x - y|^{-d} dy.$$

By translation invariance of Lebesgue measure,

$$\int_{|x-y|>\rho/2} |x-y|^{-d} dy = \int_{|z|>\rho/2} |z|^{-d} dz.$$

We compute this integral in polar coordinates in  $\mathbb{R}^{d-1}$ :

$$\int_{|z|>\rho/2} |z|^{-d} dz = \int_{\rho/2}^{\infty} r^{-d} r^{d-2} dr \int_{S^{d-2}} d\omega = C \int_{\rho/2}^{\infty} r^{-2} dr = C\rho^{-1}.$$

Therefore

$$\sum_{|x-a_Q|>\rho} |x-a_Q|^{-d} \leq C\delta_{n+1}^{-(d-1)}\rho^{-1}.$$

Substituting this estimate into the previous inequality gives

$$\sum_{Q \in \mathcal{G}_{n+1}: |x-a_Q|>\rho} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_Q))| \leq CN^{-d}\delta_{n+1}^d(\delta_{n+1}^{-(d-1)}\rho^{-1}) = CN^{-d}\delta_{n+1}\rho^{-1}.$$

The constant  $C$  depends only on a lower bound for  $\rho/\delta_{n+1}$ .  $\square$

**Corollary 3.22.** *If  $\delta_{n+1}$  is sufficiently small, then on  $\mathbb{R}^{d-1}$ ,*

$$|\nabla u_{n+1}(x) - \nabla u_n(x)| \leq CK_{n+1}e^{-\beta n}\varepsilon_{n+1}^{-d}.$$

*Proof.* Recall that

$$\nabla u_{n+1}(x) - \nabla u_n(x) = \sum_{Q' \in \mathcal{G}_{n+1}} \frac{du_n}{dn}(a_{Q'}) \nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_{Q'})).$$

**Case 1:**  $x \in Q$  for some  $Q \in \mathcal{G}_{n+1}$ .

We separate the term corresponding to  $Q' = Q$  from the rest of the sum. Using the triangle inequality we obtain

$$\begin{aligned} |\nabla u_{n+1}(x) - \nabla u_n(x)| &\leq \sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ Q' \neq Q}} \left| \frac{du_n}{dn}(a_{Q'}) \right| |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_{Q'}))| \\ &\quad + \left| \frac{du_n}{dn}(a_Q) \right| |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_Q))|. \end{aligned}$$

Since  $\delta_{n+1}$  is small and  $u_n$  is smooth, the gradient of  $u_n$  does not vary much inside cubes of size  $\delta_{n+1}$ . In particular there exists a constant  $K_{n+1}$  such that

$$\left| \frac{du_n}{dn}(a_{Q'}) \right| \leq 2K_{n+1}e^{-\beta n} \quad \text{for all } Q' \in \mathcal{G}_{n+1}.$$

Formally, Since  $u_n$  is smooth up to the boundary, the function  $g(x) = \frac{du_n}{dn}(x)$  is continuous on  $\mathbb{R}^{d-1}$ .

In particular  $g$  is uniformly continuous on the compact set  $Q(1)$ . Hence for every  $\eta > 0$  there exists  $\delta > 0$  such that

$$|x - y| < \delta \quad \Rightarrow \quad |g(x) - g(y)| < \eta.$$

Let  $Q \in \mathcal{G}_{n+1}$  be a cube with center  $a_Q$  and side length  $\delta_{n+1}$ . If  $\delta_{n+1}$  is chosen sufficiently small then for every  $x \in Q$  we have

$$|x - a_Q| \leq \frac{\sqrt{d-1}}{2} \delta_{n+1} < \delta.$$

Therefore,

$$|g(a_Q) - g(x)| < \eta,$$

which implies,

$$|g(a_Q)| \leq |g(x)| + \eta.$$

Thus

$$|g(a_Q)|^p \leq \frac{1}{|Q|} \int_Q |g(x)|^p dx + \eta^p.$$

Since  $\eta \rightarrow 0$  as  $\delta_{n+1} \rightarrow 0$ , we can choose  $\delta_{n+1}$  small enough so that

$$\eta^p \leq (2^p - 1) \frac{1}{|Q|} \int_Q |g(x)|^p dx.$$

Therefore

$$|g(a_Q)| \leq 2 \left( \frac{1}{|Q|} \int_Q |g(x)|^p dx \right)^{1/p}.$$

Finally, since  $Q \in \mathcal{G}_{n+1}$ , by definition of  $\mathcal{G}_{n+1}$  we have

$$\left( \frac{1}{|Q|} \int_Q |g(x)|^p dx \right)^{1/p} \leq K_{n+1} e^{-\beta n}.$$

Hence

$$\left| \frac{du_n}{dn}(a_Q) \right| \leq 2K_{n+1} e^{-\beta n}.$$

Therefore

$$\begin{aligned} |\nabla u_{n+1}(x) - \nabla u_n(x)| &\leq 2K_{n+1} e^{-\beta n} \left[ \sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ Q' \neq Q}} |\nabla F_{\varepsilon_{n+1}}(N \delta_{n+1}^{-1}(x - a_{Q'}))| \right. \\ &\quad \left. + |\nabla F_{\varepsilon_{n+1}}(N \delta_{n+1}^{-1}(x - a_Q))| \right]. \end{aligned}$$

The sum over  $Q' \neq Q$  is estimated using Proposition 3.21. Indeed, if  $x \in Q$  then for every  $Q' \neq Q$  one has

$$|x - a_{Q'}| \gtrsim \delta_{n+1},$$

so the hypothesis of Proposition 3.21 is satisfied. Hence

$$\sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ Q' \neq Q}} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'}))| \leq C.$$

For the remaining term we use the estimate from Proposition 3.18,

$$|\nabla F_{\varepsilon_{n+1}}(y)| \leq C \min(\varepsilon_{n+1}^{-d}, |y|^{-d}).$$

Since the minimum is always bounded by  $\varepsilon_{n+1}^{-d}$ , we obtain

$$|\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q))| \leq C\varepsilon_{n+1}^{-d}.$$

Combining the previous estimates yields

$$|\nabla u_{n+1}(x) - \nabla u_n(x)| \leq 2K_{n+1}e^{-\beta n} \left( C + C\varepsilon_{n+1}^{-d} \right) \leq CK_{n+1}e^{-\beta n}\varepsilon_{n+1}^{-d}.$$

**Case 2:  $x$  does not belong to any cube  $Q \in \mathcal{G}_{n+1}$ .**

In this case the distance from  $x$  to every center  $a_Q$  is bounded below by a multiple of  $\delta_{n+1}$ ,

$$|x - a_Q| \gtrsim \delta_{n+1}.$$

Hence all terms in the sum satisfy the hypotheses of Proposition 3.21, and we obtain directly

$$\sum_{Q \in \mathcal{G}_{n+1}} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q))| \leq C.$$

Using again the bound

$$\left| \frac{du_n}{dn}(a_Q) \right| \leq 2K_{n+1}e^{-\beta n}$$

we conclude

$$|\nabla u_{n+1}(x) - \nabla u_n(x)| \leq 2K_{n+1}e^{-\beta n}C \leq CK_{n+1}e^{-\beta n}\varepsilon_{n+1}^{-d},$$

since  $\varepsilon_{n+1}^{-d} \geq 1$ . This completes the proof.  $\square$

Now we verify the induction hypothesis in the  $(n+1)$ -th step. First another technical lemma:

**Lemma 3.23.** *Let  $\gamma > 0$  be given. If  $\delta_{n+1}$  is small enough; then*

$$\sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ Q' \neq Q}} \left| \frac{du_n}{dn}(a_{Q'}) \right| \left| \nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'})) \right| \leq CN^{-d} \left| \frac{du_n}{dn}(x) \right| + \gamma. \quad (3.10)$$

*Proof.* Let  $x \in Q$  be fixed. We split the sum into two parts according to the distance between  $x$  and the centers  $a_{Q'}$ . Let  $M > 0$  be a parameter that will be chosen later. Write

$$\sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ Q' \neq Q}} = \sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ Q' \neq Q \\ |x - a_{Q'}| < M\delta_{n+1}}} + \sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ |x - a_{Q'}| > M\delta_{n+1}}} =: S_1 + S_2.$$

For the first sum we have

$$S_1 = \sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ Q' \neq Q \\ |x - a_{Q'}| < M\delta_{n+1}}} \left| \frac{du_n}{dn}(a_{Q'}) \right| \left| \nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'})) \right|.$$

We bound the first factor by the supremum over all indices satisfying  $|x - a_{Q'}| < M\delta_{n+1}$ :

$$S_1 \leq \sup_{\{|x - a_{Q'}| < M\delta_{n+1}\}} \left| \frac{du_n}{dn}(a_{Q'}) \right| \sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ Q' \neq Q}} \left| \nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'})) \right|.$$

We now apply Proposition 3.21 which gives a uniform bound for sums of the form

$$\sum_{Q'} \left| \nabla F_{\varepsilon}(N\delta^{-1}(x - a_{Q'})) \right|.$$

More precisely, taking the parameter  $\rho = \frac{1}{2}\delta_{n+1}$  in Proposition 3.21, we obtain

$$\sum_{Q'} \left| \nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'})) \right| \leq CN^{-d}.$$

Consequently,

$$S_1 \leq CN^{-d} \sup_{\{|x - a_{Q'}| < M\delta_{n+1}\}} \left| \frac{du_n}{dn}(a_{Q'}) \right|.$$

Since  $u_n$  is harmonic in the upper half-space, the function

$$x \mapsto \frac{du_n}{dn}(x)$$

is continuous. Therefore, for every  $\gamma > 0$  there exists  $\eta > 0$  such that

$$|x - y| < \eta \implies \left| \frac{du_n}{dn}(y) \right| \leq \left| \frac{du_n}{dn}(x) \right| + \gamma.$$

If  $\delta_{n+1}$  is chosen sufficiently small so that

$$M\delta_{n+1} < \eta,$$

then for every  $Q'$  satisfying  $|x - a_{Q'}| < M\delta_{n+1}$  we obtain

$$\left| \frac{du_n}{dn}(a_{Q'}) \right| \leq \left| \frac{du_n}{dn}(x) \right| + \gamma.$$

Hence

$$S_1 \leq CN^{-d} \left( \left| \frac{du_n}{dn}(x) \right| + \gamma \right).$$

We now consider the second sum

$$S_2 = \sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ |x - a_{Q'}| > M\delta_{n+1}}} \left| \frac{du_n}{dn}(a_{Q'}) \right| \left| \nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'})) \right|.$$

We first bound the derivative term uniformly:

$$\left| \frac{du_n}{dn}(a_{Q'}) \right| \leq \sup_{y \in V_{n+1}} \left| \frac{du_n}{dn}(y) \right|.$$

Therefore

$$S_2 \leq \sup_{y \in V_{n+1}} \left| \frac{du_n}{dn}(y) \right| \sum_{|x - a_{Q'}| > M\delta_{n+1}} \left| \nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'})) \right|.$$

From the estimates established earlier in the construction we know that

$$\sup_{y \in V_{n+1}} \left| \frac{du_n}{dn}(y) \right| \leq 2K_{n+1}e^{-\beta n}.$$

Thus

$$S_2 \leq 2K_{n+1}e^{-\beta n} \sum_{|x - a_{Q'}| > M\delta_{n+1}} \left| \nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_{Q'})) \right|.$$

We again apply Proposition 3.21, this time in the form of its tail estimate. Taking

$$\rho = M\delta_{n+1},$$

we obtain

$$\sum_{|x-a_{Q'}|>M\delta_{n+1}} |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_{Q'}))| \leq CM^{-1}.$$

Therefore

$$S_2 \leq CK_{n+1}e^{-\beta n}M^{-1}.$$

Since  $M$  is arbitrary, we may choose it sufficiently large so that

$$CK_{n+1}e^{-\beta n}M^{-1} < \gamma.$$

Hence

$$S_2 < \gamma.$$

Combining the estimates for  $S_1$  and  $S_2$  we obtain

$$\sum_{\substack{Q' \in \mathcal{G}_{n+1} \\ Q' \neq Q}} \left| \frac{du_n}{dn}(a_{Q'}) \right| |\nabla F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x-a_{Q'}))| \leq CN^{-d} \left| \frac{du_n}{dn}(x) \right| + \gamma,$$

which proves the desired inequality. □

**Proposition 3.24** ([2] Lemma 4). *If  $A$  is large, then*

$$\left( \int_{V_{n+1}} \left| \frac{du_{n+1}}{dn} \right|^p \right)^{1/p} < Ae^{-\beta(n+1)}.$$

*Proof.* Let  $Q \in G_{n+1}$  and denote by  $a_Q$  the center of  $Q$ . We classify the cubes into two types:

- $Q$  is said to be *type 1* if

$$\left| \frac{du_n}{dn}(a_Q) \right| > e^{-4\beta(n+1)}.$$

- $Q$  is said to be *type 2* otherwise, that is,

$$\left| \frac{du_n}{dn}(a_Q) \right| \leq e^{-4\beta(n+1)}.$$

Fix  $Q \in G_{n+1}$  and let  $x \in Q$ . By the definition of  $u_{n+1}$  we may write

$$\begin{aligned} \frac{du_{n+1}}{dn}(x) &= \frac{du_n}{dn}(a_Q) + \left[ \frac{du_n}{dn}(x) - \frac{du_n}{dn}(a_Q) \right. \\ &\quad \left. + \sum_{\substack{Q' \in G_{n+1} \\ Q' \neq Q}} \frac{du_n}{dn}(a_{Q'}) \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x - a_{Q'})) \right] \\ &\quad + \frac{du_n}{dn}(a_Q) \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x - a_Q)). \end{aligned}$$

For convenience we introduce

$$J(x) = \frac{du_n}{dn}(x) - \frac{du_n}{dn}(a_Q) + \sum_{\substack{Q' \in G_{n+1} \\ Q' \neq Q}} \frac{du_n}{dn}(a_{Q'}) \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x - a_{Q'})).$$

By inequality (3.10) and the smoothness of  $\frac{du_n}{dn}$  we have

$$|J(x)| < CN^{-d} \left| \frac{du_n}{dn}(a_Q) \right| + \gamma$$

where  $\gamma > 0$  is small. Also since  $Q$  is type 1, the error  $\gamma$  can be taken arbitrarily small in such a way that

$$|J(x)| \lesssim N^{-d} \left| \frac{du_n}{dn}(a_Q) \right|.$$

That is, we could have a problem in chose such  $\gamma$ , if  $\frac{du_n}{dn}(a_Q)$  had no control from below, but the lower bound  $e^{-4\beta(n+1)}$  prevents that. Explicitly, choose  $\gamma \ll e^{-4\beta(n+1)}$  such that  $\gamma < CN^{-d} \left| \frac{du_n}{dn}(a_Q) \right|$ .

Choosing  $N$  sufficiently large we may assume

$$CN^{-d} < cN^{-(d-1)},$$

where  $c$  is the constant appearing in the statement of Proposition 3.19. Hence

$$|J(x)| \leq cN^{-(d-1)} \left| \frac{du_n}{dn}(a_Q) \right|.$$

Using the decomposition obtained above we write

$$\frac{du_{n+1}}{dn}(x) = \frac{du_n}{dn}(a_Q) \left( 1 + \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x - a_Q)) \right) + J(x)$$

and therefore,

$$\left| \frac{du_{n+1}}{dn}(x) \right|^p \leq \left| \frac{du_n}{dn}(a_Q) \right|^p \left| 1 + \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x - a_Q)) \right|^p + |J(x)|^p.$$

Since  $cN^{-(d-1)}$  can be taken sufficiently small, integrating and applying Proposition 3.19 yields

$$\begin{aligned} \int_Q \left| \frac{du_{n+1}}{dn} \right|^p &\leq \int_Q \left| \frac{du_n}{dn}(a_Q) \right|^p \left| 1 + \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x - a_Q)) \right|^p + |J(x)|^p \\ &< \left| \frac{du_n}{dn}(a_Q) \right|^p |Q| (e^{-2\beta p} + c^p N^{-p(d-1)}) \\ &< e^{-3/2\beta p} \int_Q \left| \frac{du_n}{dn} \right|^p. \end{aligned}$$

In the last inequality, we observe its validity when  $\delta_{n+1}$  is sufficiently small and  $N$  is large enough.

Assume now that  $Q$  is of type 2. Then

$$\left| \frac{du_n}{dn}(a_Q) \right| \leq e^{-4\beta(n+1)}.$$

Using again the decomposition we obtain

$$\frac{du_{n+1}}{dn}(x) = \frac{du_n}{dn}(a_Q) \left( 1 + \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x - a_Q)) \right) + J(x).$$

From the estimate of  $J$  we have

$$|J(x)| < CN^{-d}e^{-4\beta(n+1)} + \gamma.$$

Choosing  $\gamma$  sufficiently small we obtain

$$|J(x)| < e^{-4\beta(n+1)}.$$

Taking  $p$ -powers and integrating over  $Q$  yields

$$\begin{aligned} \int_Q \left| \frac{du_{n+1}}{dn} \right|^p &\leq \left| \frac{du_n}{dn}(a_Q) \right|^p \int_Q \left| 1 + \frac{dF_{\varepsilon_{n+1}}}{dn}(N\delta_{n+1}^{-1}(x - a_Q)) \right|^p \\ &\quad + \int_Q |J(x)|^p dx. \end{aligned}$$

By Proposition 3.19,

$$\int_Q \left| 1 + \frac{dF_{\varepsilon_{n+1}}}{dn}(\cdot) \right|^p < |Q|.$$

Moreover

$$|J(x)|^p \leq e^{-4p\beta(n+1)}.$$

Therefore

$$\int_Q \left| \frac{du_{n+1}}{dn} \right|^p \leq 2e^{-4p\beta(n+1)}|Q|.$$

Since

$$V_{n+1} = \bigcup_{Q \in G_{n+1}} Q,$$

we decompose

$$V_{n+1} = \left( \bigcup_{Q \text{ type 1}} Q \right) \cup \left( \bigcup_{Q \text{ type 2}} Q \right).$$

Hence

$$\int_{V_{n+1}} \left| \frac{du_{n+1}}{dn} \right|^p = \int_{\cup(Q \text{ type 1})} \left| \frac{du_{n+1}}{dn} \right|^p + \int_{\cup(Q \text{ type 2})} \left| \frac{du_{n+1}}{dn} \right|^p.$$

Using the previous estimates we obtain

$$\int_{V_{n+1}} \left| \frac{du_{n+1}}{dn} \right|^p \leq e^{-3p\beta/2} \int_{\cup(Q \text{ type 1})} \left| \frac{du_n}{dn} \right|^p + Ce^{-4p\beta(n+1)} \sum_{Q \text{ type 2}} |Q|.$$

Using the inductive hypothesis

$$\int_{V_n} \left| \frac{du_n}{dn} \right|^p \leq A^p e^{-p\beta n},$$

we obtain

$$\int_{V_{n+1}} \left| \frac{du_{n+1}}{dn} \right|^p \leq A^p e^{-p\beta(n+1)} \left( e^{-p\beta/2} + CA^{-p} e^{-3p\beta(n+1)} \right).$$

Choosing  $A$  sufficiently large so that

$$e^{-p\beta/2} + CA^{-p} e^{-3p\beta(n+1)} < 1,$$

we conclude that

$$\int_{V_{n+1}} \left| \frac{du_{n+1}}{dn} \right|^p \leq A^p e^{-p\beta(n+1)}.$$

It is enough to choose

$$A > \left( \frac{Ce^{-3\beta}}{1 - e^{-p\beta/2}} \right)^{1/p}$$

This completes the inductive step and hence the proof.  $\square$

### 3.2.3 Proof of Theorem 3.1

**Theorem 3.1.** *If  $d \geq 3$  there exists a harmonic function  $f : \mathcal{H}^d \rightarrow \mathbb{R}$ ,  $C^1$  up to the boundary, and such that  $f$  and  $\nabla f$  vanish on a common set of the boundary, with positive measure.*

*Proof.* We begin by referencing the construction at subsection 3.2.2 with sequences  $\{K_n\}_{n \in \mathbb{N}}, \{\varepsilon_n\}_{n \in \mathbb{N}}$  such as in Lemma (3.20).

Note that, equation (3.8) together with Corollary 3.22 implies that, the sequence of functions  $\hat{u}_n$  converges in  $C^1$  norm on  $\mathcal{H}^d$ . We denote this limit by  $f$ .

We claim that  $f$  is a harmonic function on  $\mathcal{H}^d$  which is  $C^1$  up to the boundary and that,

$$\left| \left\{ x \in Q(1) \subset \mathbb{R}^{d-1} : f(x) = \frac{df}{dn}(x) = 0 \right\} \right| > 0.$$

Indeed, since each  $u_n$  is harmonic in  $\mathcal{H}^d$ , it satisfies the Mean Value Property. Let  $x \in \mathcal{H}^d$  be an arbitrary point, and let  $B(x, r)$  be a ball such that its closure  $\overline{B(x, r)} \subset \mathcal{H}^d$ . We then notice:

$$\begin{aligned} \left| f(x) - \frac{1}{|B(x, r)|} \int_{B(x, r)} f(y) dy \right| &\leq |f(x) - u_n(x)| + \left| \frac{1}{|B|} \int_B (u_n(y) - f(y)) dy \right| \\ &\leq |f(x) - u_n(x)| + \frac{1}{|B|} \int_B |u_n(y) - f(y)| dy. \end{aligned}$$

Uniform convergence of  $u_n$  to  $f$  implies that, for any  $\varepsilon$ :

$$\left| f(x) - \frac{1}{|B(x, r)|} \int_{B(x, r)} f(y) dy \right| < \varepsilon + \frac{1}{|B|} \int_B \varepsilon dy = 2\varepsilon.$$

A continuous function satisfying the Mean Value Property for every ball in its domain is  $C^\infty$ , and satisfies Laplace's equation  $\Delta f = 0$ . This is a basic fact that follows by Theorems 2 and 6 of Chapter 2 by [16]. Thus,  $f$  is harmonic in  $\mathcal{H}^d$ . The convergence in  $C^1(\overline{\mathcal{H}^d})$  guarantees that  $f$  is  $C^1$  up to the boundary.

For the measure of  $\{x \in Q(1) : f(x) = (df/dn)(x) = 0\}$  note that if for some integer  $n_0$  we have  $u_{n+1}(x) = u_n(x), \forall n \geq n_0$  then  $f(x) = 0$ . Therefore, to have  $f(x) \neq 0$  it is necessary that  $u_{n+1}(x) \neq u_n(x)$  for infinitely many  $n$ ; in particular

$$\{x \in Q(1) : f(x) \neq 0\} \subset \bigcup_n \{x \in Q(1) : u_{n+1}(x) \neq u_n(x)\},$$

in measure:

$$|\{x \in Q(1) : f(x) \neq 0\}| \leq \sum_n |\{x \in Q(1) : u_{n+1}(x) \neq u_n(x)\}|.$$

To deal with the right hand side of the inequality, recall that

$$u_{n+1}(x) - u_n(x) = \sum_{Q \in \mathcal{G}_{n+1}} \frac{du_n}{dn}(a_Q) F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q)) \frac{\delta_{n+1}}{N}.$$

Hence  $u_{n+1}(x) \neq u_n(x)$  only if for some  $Q \in \mathcal{G}_{n+1}$

$$F_{\varepsilon_{n+1}}(N\delta_{n+1}^{-1}(x - a_Q)) \neq 0.$$

From Proposition 3.18 we know that

$$\text{supp } F_\varepsilon \subset D(0, \varepsilon^{1/2}),$$

and therefore

$$F_{\varepsilon_{n+1}}(y) \neq 0 \quad \Rightarrow \quad |y| \leq \varepsilon_{n+1}^{1/2}.$$

Substituting  $y = N\delta_{n+1}^{-1}(x - a_Q)$  gives

$$|N\delta_{n+1}^{-1}(x - a_Q)| \leq \varepsilon_{n+1}^{1/2},$$

which implies

$$|x - a_Q| \leq \frac{\delta_{n+1}}{N} \varepsilon_{n+1}^{1/2}.$$

Thus the set  $\{x : u_{n+1}(x) \neq u_n(x)\}$  is contained in the union of discs

$$D\left(a_Q, \frac{\delta_{n+1}}{N} \varepsilon_{n+1}^{1/2}\right), \quad Q \in \mathcal{G}_{n+1}.$$

The cubes in  $\mathcal{G}_{n+1}$  have side length  $\delta_{n+1}$  and lie in  $\mathbb{R}^{d-1}$ . Consequently the number of cubes intersecting  $Q(1)$  is bounded by

$$C \delta_{n+1}^{-(d-1)}.$$

Therefore  $\{u_{n+1} \neq u_n\}$  is contained in a union of at most

$$C \delta_{n+1}^{-(d-1)}$$

discs.

We now estimate the measure of each disc. In  $\mathbb{R}^{d-1}$  the measure of a ball of radius  $r$  satisfies

$$|D(0, r)| \leq C r^{d-1}.$$

Taking

$$r = \frac{\delta_{n+1}}{N} \varepsilon_{n+1}^{1/2}$$

we obtain

$$|D(a_Q, r)| \leq C(\delta_{n+1}\varepsilon_{n+1}^{1/2})^{d-1}.$$

Combining the estimate for the number of discs with the estimate for their size yields

$$\begin{aligned} |\{x \in Q(1) : u_{n+1}(x) \neq u_n(x)\}| &\leq C\delta_{n+1}^{-(d-1)}(\delta_{n+1}\varepsilon_{n+1}^{1/2})^{d-1} \\ &= C\delta_{n+1}^{-(d-1)}\delta_{n+1}^{d-1}\varepsilon_{n+1}^{(d-1)/2} \\ &\leq C\varepsilon_{n+1}^{(d-1)/2}. \end{aligned}$$

Finally, inserting this bound into the previous estimate gives

$$|\{x \in Q(1) : f(x) \neq 0\}| \leq C \sum_n \varepsilon_{n+1}^{(d-1)/2}.$$

Recall that

$$V_n = \bigcup_{Q \in \mathcal{G}_n} Q,$$

where  $\mathcal{G}_n$  denotes the family of cubes selected at step  $n$  of the construction.

From the inductive construction we know that, if  $\frac{df}{dn}(x) \neq 0$ , then  $x$  cannot belong to all the sets  $V_n$ . Hence there exists some index  $n$  such that

$$x \in V_n \setminus V_{n+1}.$$

Therefore

$$\left\{x \in Q(1) : \frac{d\hat{f}}{dn}(x) \neq 0\right\} \subset \bigcup_n (V_n \setminus V_{n+1}).$$

in measure:

$$\left|\left\{x \in Q(1) : \frac{d\hat{f}}{dn}(x) \neq 0\right\}\right| \leq \sum_n |V_n \setminus V_{n+1}|.$$

Let  $Q \in \mathcal{G}_n$  and suppose that  $Q \notin \mathcal{G}_{n+1}$  but  $Q \subset Q'$  for some cube  $Q' \in \mathcal{G}_n$ . By the stopping condition used in the construction we have

$$|Q|^{-1} \int_Q \left| \frac{du_n}{dn} \right|^p > K_{n+1}^p e^{-p\beta n}.$$

Equivalently,

$$|Q| < K_{n+1}^{-p} e^{p\beta n} \int_Q \left| \frac{du_n}{dn} \right|^p.$$

Since

$$V_n \setminus V_{n+1} = \bigcup_{\substack{Q \in \mathcal{G}_n \\ Q \notin \mathcal{G}_{n+1}}} Q,$$

and the cubes in  $\mathcal{G}_n$  are pairwise disjoint, we obtain

$$\begin{aligned} |V_n \setminus V_{n+1}| &= \sum_{\substack{Q \in \mathcal{G}_n \\ Q \notin \mathcal{G}_{n+1}}} |Q| \\ &\leq \sum_{\substack{Q \in \mathcal{G}_n \\ Q \notin \mathcal{G}_{n+1}}} K_{n+1}^{-p} e^{p\beta n} \int_Q \left| \frac{du_n}{dn} \right|^p. \end{aligned}$$

Factoring out the constant gives

$$|V_n \setminus V_{n+1}| \leq K_{n+1}^{-p} e^{p\beta n} \sum_{\substack{Q \in \mathcal{G}_n \\ Q \notin \mathcal{G}_{n+1}}} \int_Q \left| \frac{du_n}{dn} \right|^p.$$

Since the cubes are disjoint and contained in  $V_n$ , we have

$$\sum_Q \int_Q \left| \frac{du_n}{dn} \right|^p \leq \int_{V_n} \left| \frac{du_n}{dn} \right|^p.$$

Hence

$$|V_n \setminus V_{n+1}| \leq K_{n+1}^{-p} e^{p\beta n} \int_{V_n} \left| \frac{du_n}{dn} \right|^p.$$

Finally, by the inductive estimate established earlier in the construction we have

$$\int_{V_n} \left| \frac{du_n}{dn} \right|^p \leq A^p e^{-p\beta n}.$$

Substituting this bound yields

$$|V_n \setminus V_{n+1}| \leq A^p K_{n+1}^{-p}.$$

That is,

$$\begin{aligned} |\{(df/dn) \neq 0 \text{ or } f \neq 0\}| &\leq |\{x \in Q(1) : (df/dn)(x) \neq 0\}| + |\{x \in Q(1) : f(x) \neq 0\}| \\ &\leq C \sum \varepsilon_{n+1}^{(d-1)/2} + A^p \sum K_{n+1}^{-p} \\ &\quad (\text{Equation (3.9)}) < 1, \end{aligned}$$

therefore,  $Q(1) \cap \{(df/dn) = 0\} \cap \{f = 0\}$  has positive measure.

□

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